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Organization of the
United Nations



GLOBAL SOIL ORGANIC CARBON Map

GSOCmap v.1.6

TECHNICAL REPORT

Global Soil Organic Carbon map (GSOCmap) v1.6 Technical report

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Contents

Foreword	xv
Abbreviations and acronyms	xvii
Contributors	xix
1 Background	1
1.1 The importance of soil organic carbon	1
1.2 Objectives of soil organic carbon mapping	2
1.3 Data policy	4
1.3.1 Data sharing principles	4
1.3.2 Ownership, data rights and citation	5
2 GSP capacity development programme	7
2.1 Training courses on digital soil mapping	7
2.2 GSP remote support platform	9
2.3 Soil organic carbon mapping cookbook	9
2.4 GSP and ISRIC environmental covariates data repository	10

3	Product specifications	18
3.1	Generic target specification	18
3.2	Metadata specifications	19
4	Data collection and processing	21
4.1	Different scenarios of country driven action	21
4.1.1	Delivery of the maps produced by the countries	22
4.1.2	Joint efforts	23
4.1.3	GSP gapfilling	24
4.2	Data processing and compilation of the GSOCmap	25
5	Metadata	26
5.1	Sampling density	26
5.2	Temporal dimension	27
5.3	Soil organic carbon	28
5.4	Bulk density	29
5.5	Coarse fragments	31
5.6	Mapping methods	31
6	Results	34
6.1	Total global soil organic carbon stock	34
6.1.1	Statistics for countries (GSOCmap V 1.6.0)	35
6.1.2	Statistics for climate zones, land use and soil types	36
6.2	Validation, uncertainty and comparison with existing products	40
6.2.1	Validation of the GSOCmap using available data	40
6.2.2	Uncertainty analysis	46
6.3	GSOCmap web services	47

7 Conclusions and the way forward	50
Versioning system	51
References	52
Annex A: Metadata questionnaire	58
A1: Source data	58
A2: Analysis methods	58
A3: Mapping	59
A4: Contact details	59
Annex B: Changelog file of the GSOCmap	60
Annex C: Example scripts used in GSP gapfilling	67
C1: Data preparation for soil profiles	67
C2: Mixing covariates and soil points data	70
C3: Fitting a RK model to predict the OCS	71
C4: Fitting a random forest model to predict the OCS	74
C5: Fitting a svm model to predict the OCS	75
Annex D: GSOCmap submission overview	76
D1: Afghanistan	76
D2: Albania	77
D3: Algeria	77
D4: Andorra	78
D5: Angola	79
D6: Antigua and Barbuda	80
D7: Argentina	81

D8: Armenia	82
D9: Australia	82
D10: Austria	83
D11: Azerbaijan	84
D12: Bahamas	85
D13: Bahrain	86
D14: Bangladesh	86
D15: Barbados	87
D16: Belarus	87
D17: Belgium	88
D18: Belize	90
D19: Benin	91
D20: Bhutan	92
D21: Bolivia (Plurinational State of)	93
D22: Bosnia and Herzegovina	94
D23: Botswana	95
D24: Brazil	96
D25: Brunei Darussalam	97
D26: Bulgaria	97
D27: Burkina Faso	98
D28: Burundi	99
D29: Cambodia	100
D30: Cameroon	101
D31: Canada	102
D32: Cape Verde	103
D33: Central African Republic	103

D34: Chad	104
D35: Chile	105
D36: China	106
D37: Colombia	107
D38: Comoros	108
D39: Congo	109
D40: Cook Islands	110
D41: Costa Rica	110
D42: Côte d'Ivoire	111
D43: Croatia	112
D44: Cuba	112
D45: Cyprus	113
D46: Czech Republic	114
D47: Democratic People's Republic of Korea	115
D48: Democratic Republic of the Congo	115
D49: Denmark	116
D50: Djibouti	117
D51: Dominica	118
D52: Dominican Republic	118
D53: Ecuador	119
D54: Egypt	120
D55: El Salvador	121
D56: Equatorial Guinea	122
D57: Eritrea	123
D58: Estonia	124
D59: Eswatini	125

D60: Ethiopia	126
D61: Faroe Islands	127
D62: Fiji	128
D63: Finland	128
D64: France	129
D65: Gabon	130
D66: Gambia	131
D67: Georgia	132
D68: Germany	132
D69: Ghana	133
D70: Greece	134
D71: Grenada	135
D72: Guatemala	136
D73: Guinea	137
D74: Guinea-Bissau	138
D75: Guyana	139
D76: Haiti	140
D77: Honduras	140
D78: Hungary	142
D79: Iceland	142
D80: India	143
D81: Indonesia	144
D82: Iran (Islamic Republic of)	145
D83: Iraq	146
D84: Ireland	147
D85: Israel	148

D86: Italy	149
D87: Jamaica	150
D88: Japan	151
D89: Jordan	152
D90: Kazakhstan	153
D91: Kenya	154
D92: Kiribati	155
D93: Kuwait	155
D94: Kyrgyzstan	156
D95: Lao People's Democratic Republic	157
D96: Latvia	158
D97: Lebanon	159
D98: Lesotho	160
D99: Liberia	161
D100: Libya	162
D101: Lithuania	163
D102: Luxembourg	164
D103: Madagascar	165
D104: Malawi	166
D105: Malaysia	166
D106: Maldives	167
D107: Mali	167
D108: Malta	168
D109: Marshall Islands	169
D110: Mauritania	170
D111: Mauritius	171

D112: Mexico	171
D113: Micronesia (Federated States of)	173
D114: Monaco	173
D115: Mongolia	174
D116: Montenegro	175
D117: Morocco	175
D118: Mozambique	176
D119: Myanmar	177
D120: Namibia	178
D121: Nauru	179
D122: Nepal	179
D123: Netherlands	180
D124: New Zealand	181
D125: Nicaragua	182
D126: Niger	182
D127: Nigeria	183
D128: Niue	184
D129: North Macedonia	185
D130: Norway	186
D131: Oman	186
D132: Pakistan	187
D133: Palau	188
D135: Panama	189
D136: Papua New Guinea	190
D137: Paraguay	190
D138: Peru	191

D139: Philippines	192
D140: Poland	193
D141: Portugal	194
D142: Qatar	195
D143: Republic of Korea	196
D144: Republic of Moldova	196
D145: Romania	197
D146: Russian Federation	198
D147: Rwanda	199
D148: Saint Kitts and Nevis	200
D149: Saint Lucia	201
D150: Saint Vincent and the Grenadines	201
D151: Samoa	202
D153: Sao Tome e Principe	202
D154: Saudi Arabia	203
D155: Senegal	204
D156: Serbia	204
D157: Seychelles	205
D158: Sierra Leone	205
D159: Singapore	206
D160: Slovakia	207
D161: Slovenia	208
D162: Solomon Islands	208
D163: Somalia	209
D164: South Africa	210
D165: South Sudan	210

D166: Spain	211
D167: Sri Lanka	212
D168: Sudan	213
D169: Suriname	214
D170: Sweden	215
D171: Switzerland	216
D172: Syrian Arab Republic	217
D173: Tajikistan	218
D174: United Republic of Tanzania	219
D175: Thailand	219
D176: Timor-Leste	220
D177: Togo	221
D178: Tokelau	222
D179: Tonga	222
D180: Trinidad and Tobago	222
D181: Tunisia	223
D182: Turkey	224
D183: Turkmenistan	225
D184: Tuvalu	226
D185: Uganda	227
D186: Ukraine	228
D187: United Arab Emirates	229
D188: United Kingdom of Great Britain and Northern Ireland	230
D189: United States of America	230
D190: Uruguay	231
D191: Uzbekistan	232

D192: Vanuatu	233
D193: Venezuela (Bolivarian Republic of)	234
D194: Viet Nam	235
D195: Yemen	235
D196: Zambia	236
D197: Zimbabwe	237

Figures

2.1	Density of point data (per country)	8
4.1	Country driven action depending on national capacity	22
5.1	Density of point data (per country)	27
5.2	Sampling period	28
5.3	SOC analysis methods	29
5.4	Bulk density analysis methods)	30
5.5	Coarse fragments methods)	31
5.6	Mapping methods	32
5.7	Fragment of the GSOCmap with different techniques used for mapping at country level	33
6.1	GSOCmap version 1.6.0.	34
6.2	Soil organic carbon stocks per IPCC climate region calculated from the GSOCmap v1.6.0 and aboveground organic carbon from IPCC (2008)	37
6.3	Comparison between different estimates of global SOC stocks in topsoils	41
6.4	Locations of points in the validation dataset based on publicly available data	42

6.5	Change vector maps. Derived from a standardized confusion matrix. The map in A shows the changes from GSOCmap to the HWSD, B from the GSOCmap to SoilGrids and C from HWSD to SoilGrids.	45
6.6	Map of standard deviations	47
6.7	Accessing the GSOCmap on the web	48
6.8	GSOCmap web services architecture	49

Tables

- 2.1 Covariates. 12
- 4.1 Sources of the country maps included in the GSOCmap v1.6 . . . 24
- 6.1 Global SOC estimates. 35
- 6.2 SOC stocks for selected countries. 35
- 6.3 SOC stocks by IPCC climate region. 36
- 6.4 SOC stocks by land cover. 38
- 6.5 SOC stocks by soil type. 39
- 6.6 Comparison of validation criteria between the three SOC products based on international publicly available datasets 43

Foreword

This document presents the technical details of the first-ever country-driven Global Soil Organic Carbon Map (GSOCmap). This map allows the estimation of Soil Organic Carbon (SOC) stocks from 0 to 30 cm. It represents a key contribution to the Sustainable Development Goal (SDG) indicator 15.3.1, which defines the area of degraded land. The novelty of this map is the fact that it is the first Global SOC stocks assessment which is produced through a participatory approach. Supported by the GSP-Secretariat, countries developed their capacities and stepped up efforts to compile or collect all available soil information at the national level. The 5th GSP Plenary Assembly, in June 2017, approved the decision of Members jointly developing a GSOCmap as a baseline for the amount and distribution of SOC in soils. This map is part of the process of building a Global Soil Information System (GloSIS) under the Soil Information and Data (SID) area of work of the GSP. This considerable effort, which led to the launch of the GSOCmap on World Soil Day 2017 (5 December), is paving the way to the establishment of national soil information systems and represents the first step toward introducing a soil monitoring program.

This technical report is aimed at scientists. It provides guidance on the process that led to the establishment of the GSOCmap and on how to use it. The map provides users with useful information to monitor the soil condition, identify degraded areas, set restoration targets, explore SOC sequestration potentials, support the greenhouse gas emission reporting under the UNFCCC. It also provides users with crucial information that is needed to make evidence-based decisions to mitigate and adapt to climate change.

We can expect that the extensive data content of more than 1 million sampling points resulting from country contributions and the interactive nature of the GSOCmap will greatly assist in the process of building a global soil information

system, which in turn will help contribute in the achievement of the SDGs. The contributing institutions from the participating countries which submitted their maps or data are listed in the Appendix. This technical report is a companion report to the GSOCmap V1.6.0. It presents methodologies and process of compiling the Global Soil organic carbon Map.

Abbreviations and acronyms

BD Bulk Density

CO₂ Carbon dioxide

CRF Coarse fragments

DM Dry matter

DSM Digital soil mapping

GAUL Global Administrative Unit Layers

GHG Greenhouse gas

GSOCmap Global Soil Organic Carbon Map

GSOCseq Global Soil Organic Carbon Sequestration Potential Map

GSP Global Soil Partnership

HWSD Harmonized World Soil Database

ISCN International Soil Carbon Network

INSII International Network of Soil Information Institutions

IPBES Intergovernmental Platform on Biodiversity and Ecosystem Services

IPCC Intergovernmental Panel on Climate Change

IPR Intellectual Property Rights

ITPS Intergovernmental Technical Panel on Soils

LDN Land Degradation Neutrality

NDVI Normalized difference in vegetation index

NPP Net Primary Production

P4WG Pillar 4 Working Group

QA/QC Quality Assurance/Quality Check

RMSE Root mean square error

SDF Soil Data Facility

SDG Sustainable Development Goals

SISLAC Latin America and the Caribbean's Soil Information System

SOC Soil organic carbon

SOM Soil organic matter

SPADE/M Soil Profile Analytical Database of Europe of Measured Parameters

SWRS Status of World's Soil Resources

UNCCD United Nations Convention to Combat Desertification

WFS Web Feature Service

WoSIS World Soil Information Service

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Chapter 1

Background

1.1 The importance of soil organic carbon

Soil organic carbon (SOC) is the main component of soil organic matter (SOM) and is a crucial contributor to food production, mitigation and adaption to climate change, and the achievement of the Sustainable Development Goals (SDGs). SOC affects most of the processes relevant to soil functions and food production. A high SOM, and therefore SOC content provides plants with the nutrients and water they need by increasing soil fertility and water availability, which in turn improve food productivity. SOC has also long been used as an indicator of soil health, due to its capacity to improve soil structural stability, which affects porosity, aeration and water filtration capacities to supply clean water. However, SOC mineralization can be an important source of greenhouse gas (GHG) emissions. This means that changing SOM (and hence SOC) not only changes the provision of ecosystem services required for crop production, but also affects soils' capacity to buffer against environmental changes, as it regulates the resilience agricultural systems to climate change.

SOC has received great attention during the development of the greenhouse gas (GHG) reporting programme of the Intergovernmental Panel on Climate Change (IPCC) since the mid-nineties. This was done to address the contribution of intensive land management and the vast amount of degraded land to greenhouse gas emissions, since these have caused tremendous historic losses of SOC, resulting in high potentials for future carbon storage. Recently, an

increasing number of authors have stressed the crucial role of healthy soils, with soil carbon being the most important indicator for food security and resilience against climate change. This has led to above and below ground carbon (SOC) becoming sub-indicators for the Sustainable Development Goals (SDG) target 15.3.1 (Proportion of land that is degraded over total land area).

The Status of the World's Soil Resources (SWRS) report highlights that, although more carbon is stored in the soil than in the atmosphere and plant life combined, a large portion (33 percent) of the world's soils are degraded, which has led to a major loss of global SOC reserves. The reversal of soil degradation through the buildup of SOM and the sustainable management of soils therefore offers large potential to contribute to climate change mitigation by sequestering atmospheric carbon into the soil. This emphasizes that soil can be a double-edged sword when it comes to carbon fluxes, as it can either be a net sink or a net source of GHGs depending on soil management practices.

Despite the current focus on SOC, knowledge about SOC baselines and changes, and the detection of vulnerable hot spots for SOC losses and gains under climate change and changed land management are still fairly limited. Accurate SOC baselines are still needed for many countries, and estimates about the role of soils in the global carbon cycle are currently only based on rough estimates, which results in large uncertainties. Global SOC estimates exist, but there is high variability in reported values among authors, caused by the diversity of different data sources and methodologies used to calculate and measure these estimates (Henry *et al.*, 2009; Köchy *et al.*, 2015).

1.2 Objectives of soil organic carbon mapping

The Intergovernmental Technical Panel on Soils (ITPS) and the GSP Secretariat were asked by the Science-Policy Interface (SPI) of the United Nations Convention to Combat Desertification (UNCCD) to share information about the possible pathways to support the SDG 15.3.1 indicator on SOC. During the fifth Session of the ITPS held during March 2016, collaboration between the ITPS and the SPI of the UNCCD, the Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES), and the Intergovernmental Panel on Climate Change (IPCC) was discussed. The GSP/ITPS were requested to conduct a global SOC assessment based on country-level spatial soil data sets, combined into a new global SOC map. This task would directly relate to SDG 15.3.1, and

would also support the endorsed metrics for the assessment of land degradation neutrality (LDN) (ITPS & FAO, 2015). The preparation of the GSOCmap was discussed and supported to the fourth and fifth GSP Plenary Assembly (FAO & GSP, 2016; FAO & GSP, 2017a).

As it was approved by the decision of the 5th GSP Plenary Assembly, June 2017 (FAO & GSP, 2017), GSP members agreed to jointly develop a global SOC map as the zero status for the amount and distribution of SOC in soils around the world. This map was developed following the general GSP principle of being a country-driven initiative. It is a part of the process to build a Global Soil Information System under GSP Pillar 4 (Enhance the quantity and quality of soil data and information: data collection (generation), analysis, validation, reporting, monitoring and integration with other disciplines).

The development of a global soil organic carbon map using a country-driven approach provides and builds on synergies with ongoing and new reporting needs, data sharing obligations, and therefore benefits activities at national, regional and global levels. Particularly to:

- enable training for countries in need of technical support (e.g. regarding the collection, statistical evaluation and modeling of SOC data);
- develop data infrastructure to update the SWRS report on SOC through a country-driven baseline, and initiate future assessments of SOC changes;
- support national GHG reporting: develop a valid, measurement-based inventory of reference SOC stocks for IPCC-Tier 2 assessments;
- further utilize SOC mapping to estimate the soil carbon sequestration potentials (e.g. through modeling) and the vulnerability of soil functions under climate change (with SOC as an indicator);
- contribute to the Sustainable Development Goals: by developing national SDG-15.3.1 Tier 3 data for the sub-indicator of soil carbon; and
- conduct harmonized assessments at different levels of action: GSP regional soil partnerships, FAO regional and country offices, national soil information institutions (GSP Pillar 4 INSII), national statistics offices (already involved with FAOSTAT), and GEOSS design principles for global data layers.

1.3 Data policy

1.3.1 Data sharing principles

The GSP Data Policy has been endorsed by partners of the Global Soil Partnership during the 5th GSP Plenary Assembly in June 2017 (GSP & FAO, 2017) in order to promote and govern soil data sharing for data products including GSOCmap contributions, and considering harmonization and interoperability requirements.

The GSP data policy aims to ensure that:

- every existing ownership right to shared soil data are respected;
- the specific level of access and the conditions for data sharing are clearly specified;
- the ownership of each dataset and web service are properly acknowledged and well-referenced; and
- the data owners are protected from any liability arising from the use of their original and/or derived data.

It is recommended that data owners comply with the following open data principles:

1. Accessibility: the data shall be divulged through the Internet (web services).
2. Availability: the data is presented in a convenient, platform-independent, and in line with standard formats (e.g. web feature service WFS).
3. License: the formal concession of the usage and access rights over the data shared.
4. Cost: data shall be shared free of cost, or at no more than a reasonable reproduction cost, preferably by downloading it from the Internet.
5. Re-use and Redistribution: data must be provided and licensed under terms that permit its reuse and redistribution, including intermixing with other datasets.

6. Global benefit: any user must be able to access, use and redistribute data of the Global Soil Information System. However, inherited restrictions by national data policies shall be accepted.
7. Metadata: data describing the products of the Global Soil Information System will by default be open for access.

The data shared by the countries shall contain the relevant soil information representative for the area portrayed. The shared datasets contain the best available information for a given area and topic, however, they are subject to potential restrictions based on the institutions' or countries' data policy.

The data shared by the countries should be quality controlled which means that the data have been technically evaluated to ensure data integrity, correctness, and completeness; errors and omissions are identified and, if possible, addressed.

1.3.2 Ownership, data rights and citation

In the case of original data, the rightful data owner keeps full ownership of it. All intellectual property rights (IPR) and copyrights pertaining to the data owner remain intact and are respected by the soil data facility (SDF) host. All data providers must communicate to the SDI host their IPR and data use policies. Thus, the ownership of all data made available through the GSP soil portal need to be clearly specified. This is an important prerequisite to allow this data to be accessible through the soil SDF.

In the case of derived data, the deriving institution becomes the rightful owner. However, all original data must be accredited and correctly cited. According to the Pillar 4 Implementation Plan, each global-level derived GSP data product will be quality-assured by the Pillar 4 Working Group. This includes agreements about the correct citation.

The data owner shall ensure that the data shared can be used and interpreted by the authorized users in general; this includes providing the proper citations, as well as providing information over the ownership of such data for acknowledgement purposes. Users shall acknowledge the source of data provided through the Global Soil Information System.

All providers of original data (data owners) are responsible to define and clarify the IPR and licensing. Any user of this data, such as the SDF host, has to respect the national data policies and/or licensing involved with the retrieval of

the respective web services. In the case of data provided to the central repository, a bilateral agreement/license may be required (between the national data owner and SDF host), depending on and in conformity with national rules. More information about the data policy can be accessed at GSP & FAO (2017).

Chapter 2

GSP capacity development programme

2.1 Training courses on digital soil mapping

Considering the request from partners to support them by providing training on state of the art techniques for SOC mapping, the Secretariat designed a capacity development programme following an on-the-job training model. The aim of the GSP capacity development program has been to introduce recent concepts and techniques of digital soil mapping (DSM) to soil experts who work at national soil science institutes in soil mapping related activities. The impact of the trainings should be reflected on developing and updating national and regional soil information systems.



Figure 2.1: GSP capacity development programme: Training locations and participated countries. Source: UN Geospatial, 2020, modified by authors.

In order to support national capacities on digital soil organic carbon mapping, DSM workshops were organized by the GSP and the regional soil partnerships. The training workshops were already part of the GSP capacity development programme before the launch of the GSOCmap project (2012–2016). In 2017, the training focused more on digital soil organic carbon mapping to support countries with their GSOCmap contributions. After the launch of the GSOCmap project, additional training sessions were organized in different regions and eventually the capacity development programme was able to reach 105 countries and 60 percent of the area coverage (see Figure 2.1).

The contents of the workshops included: introduction to R; preparing spatial covariates using SAGA GIS; correlation analysis; regression–kriging; random-Forest; support vector machines; uncertainties and validation. By the end of the training courses, participants were able to collect and rescue soil legacy data, compile and harmonize soil data for DSM applications, implement DSM, produce soil property maps and their uncertainties, and develop accurate digital soil maps for updating their national soil information systems.

2.2 GSP remote support platform

There was a systematic support of the GSP Secretariat to assist and provide technical support to soil experts after the training sessions (e.g., phone, video conferencing, email exchange). The post-training support process allowed the GSP Secretariat to address any questions, doubts, or problems that countries faced.

2.3 Soil organic carbon mapping cookbook

The soil organic carbon mapping cookbook (Yigini *et al.*, 2018) has been developed by the ITPS and GSP Secretariat to provide generic methodologies and the technical steps for producing a SOC map. This includes step-by-step guidance for developing 1 km grids for SOC stocks, as well as for the preparation of local soil data, the compilation and pre-processing of ancillary spatial data sets, mapping methodologies, and uncertainty assessments.

Guidance is mainly specific to soil organic carbon data, but also contains many generic sections on soil grid development due to its relevance for other soil properties. The main focus of the guidance is on the mapping of SOC stocks in the GSOCmap and as such the cookbook supplements the GSP Guidelines for sharing national data/information to compile a Global Soil Organic Carbon (GSOC) map. It provides technical guidance to:

- setting up the needed software environment;
- preparing ground data for soil organic carbon modelling;
- calculating SOC stocks from local samples to a target depth of 30 cm;
- preparing spatial covariates for mapping;
- choosing and applying the best suitable mapping methodology; and
- evaluating the results and the outputs and providing guidance on validation and uncertainty assessments.

2.4 GSP and ISRIC environmental covariates data repository

A set of standardized national environmental covariates for digital soil mapping were provided by ISRIC World Soil Information (Table 2.1). The data can be accessed through ftp, all necessary credentials have been provided to the countries. The provided data sets fall within the following thematic fields:

- geomorphometry i.e. digital elevation models and derived land surface parameters and objects;
- spectral and multispectral remote sensing imagery and derived parameters;
- climatic and meteorological covariates;
- land cover/land use information; and
- parent material and soil-unit maps.

This data repository contains GIS raster layers of various biophysical earth surface properties for each territory in the world. These layers can, for example, be used as covariates in a digital soil mapping exercise. The territories and their boundaries are obtained from the Global Administrative Unit Layers (GAUL) dataset ¹

Each folder contains three sub-folders:

- covs: GIS layers of various biophysical earth surface properties; and
- mask: an ‘empty’ grid file of the territory with territory boundary according to GAUL. This grid can for instance be used as a mapping mask.

Data Specifications

File format: GeoTiff

Coordinate system: WGS84, latitude-longitude in decimal degrees

Spatial resolution: 1 km

¹<http://www.fao.org/geonetwork/srv/en/metadata.show?id=12691>

Data Access: <https://files.isric.org/projects/gsp/> (user: gsp, pwd: gspisric)

Licence and Acknowledgement

The GIS layers can be freely used under the condition that proper credit should be given to the original data source in each publication or product derived from these layers. Licences, data sources, and data citations are indicated in the data description table.

Table 2.1: Covariates.

Name	Series	Attribute
DEMENV5	DEM-parameters	Land surface elevation
SLPMRG5	DEM-parameters	Terrain slope
CRVMRG5	DEM-parameters	Downslope Curvature
CRUMRG5	DEM-parameters	Local upslope Curvature
CRDMRG5	DEM-parameters	Local downslope Curvature
VBFMRG5	DEM-parameters	Multiresolution Index of Valley Bottom Flatness
DVMMRG5	DEM-parameters	Deviation from Mean Value (surface roughness) x9
DV2MRG5	DEM-parameters	Deviation from Mean Value (surface roughness) x13 VDPMRG5
DEM-parameters	Valley depth	
NEGMRG5	DEM-parameters	Negative Topographic Openness
POSMRG5	DEM-parameters	Positive Topographic Openness
MRNMRG5	DEM-parameters	Melton Ruggedness Number
TPIMRG5	DEM-parameters	Topographic Position Index
TWIMRG5	DEM-parameters	SAGA Wetness Index
EX1MOD5	MOD13Q1	Mean monthly MODIS EVI JanFeb
EX2MOD5	MOD13Q1	Mean monthly MODIS EVI MarApr
EX3MOD5	MOD13Q1	Mean monthly MODIS EVI MayJun
EX4MOD5	MOD13Q1	Mean monthly MODIS EVI JulAug
EX5MOD5	MOD13Q1	Mean monthly MODIS EVI SepOct
EX6MOD5	MOD13Q1	Mean monthly MODIS EVI NovDec
ES1MOD5	MOD13Q1	SD monthly MODIS EVI JanFeb
ES2MOD5	MOD13Q1	SD monthly MODIS EVI MarApr
ES3MOD5	MOD13Q1	SD monthly MODIS EVI MayJun
ES4MOD5	MOD13Q1	SD monthly MODIS EVI JulAug
ES5MOD5	MOD13Q1	SD monthly MODIS EVI SepOct
ES6MOD5	MOD13Q2	SD monthly MODIS EVI NovDec
I01MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Jan

Table 2.1: Covariates. (*continued*)

Name	Series	Attribute
I02MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Feb
I03MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Mar
I04MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Apr
I05MOD4	MCD43A4	Mean monthly MODIS NIR band 4 May
I06MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Jun
I07MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Jul
I08MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Aug
I09MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Sep
I10MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Oct
I11MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Nov
I12MOD4	MCD43A4	Mean monthly MODIS NIR band 4 Dec
M01MOD4	MCD43A4	Mean monthly MODIS MIR band 7 Jan
M02MOD4	MCD43A4	Mean monthly MODIS MIR band 7 Feb
M03MOD4	MCD43A4	Mean monthly MODIS MIR band 7 Mar
M04MOD4	MCD43A4	Mean monthly MODIS MIR band 7 Apr
M05MOD4	MCD43A4	Mean monthly MODIS MIR band 7 May
M06MOD4	MCD43A4	Mean monthly MODIS MIR band

Table 2.1: Covariates. (*continued*)

Name	Series	Attribute
M07MOD4	MCD43A4	7 Jun Mean monthly MODIS MIR band
M08MOD4	MCD43A4	7 Jul Mean monthly MODIS MIR band
M09MOD4	MCD43A4	7 Aug Mean monthly MODIS MIR band
M10MOD4	MCD43A4	7 Sep Mean monthly MODIS MIR band
M11MOD4	MCD43A4	7 Oct Mean monthly MODIS MIR band
M12MOD4	MCD43A4	7 Nov Mean monthly MODIS MIR band
T01MOD3	MOD11A2	7 Dec Mean monthly MODIS LST (daytime) Jan
T02MOD3	MOD11A2	Mean monthly MODIS LST (daytime) Feb
T03MOD3	MOD11A2	Mean monthly MODIS LST (daytime) Mar
T04MOD3	MOD11A2	Mean monthly MODIS LST (daytime) Apr
T05MOD3	MOD11A2	Mean monthly MODIS LST (daytime) May
T06MOD3	MOD11A2	Mean monthly MODIS LST (daytime) Jun
T07MOD3	MOD11A2	Mean monthly MODIS LST (daytime) Jul
T08MOD3	MOD11A2	Mean monthly MODIS LST (daytime) Aug
T09MOD3	MOD11A2	Mean monthly MODIS LST (daytime) Sep
T10MOD3	MOD11A2	Mean monthly MODIS LST

Table 2.1: Covariates. (*continued*)

Name	Series	Attribute
T11MOD3	MOD11A2	(daytime) Oct Mean monthly MODIS LST (daytime) Nov
T12MOD3	MOD11A2	Mean monthly MODIS LST (daytime) Dec
N01MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Jan
N02MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Feb
N03MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Mar
N04MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Apr
N05MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) May
N06MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Jun
N07MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Jul
N08MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Aug
N09MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Sep
N10MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Oct
N11MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Nov
N12MOD3	MOD11A2	Mean monthly MODIS LST (nighttime) Dec
T01MSD3	MOD11A2	SD monthly MODIS LST (daytime) Jan
T02MSD3	MOD11A2	SD monthly MODIS LST (daytime) Feb

Table 2.1: Covariates. (*continued*)

Name	Series	Attribute
T03MSD3	MOD11A2	SD monthly MODIS LST (daytime) Mar
T04MSD3	MOD11A2	SD monthly MODIS LST (daytime) Apr
T05MSD3	MOD11A2	SD monthly MODIS LST (daytime) May
T06MSD3	MOD11A2	SD monthly MODIS LST (daytime) Jun
T07MSD3	MOD11A2	SD monthly MODIS LST (daytime) Jul
T08MSD3	MOD11A2	SD monthly MODIS LST (daytime) Aug
T09MSD3	MOD11A2	SD monthly MODIS LST (daytime) Sep
T10MSD3	MOD11A2	SD monthly MODIS LST (daytime) Oct
T11MSD3	MOD11A2	SD monthly MODIS LST (daytime) Nov
T12MSD3	MOD11A2	SD monthly MODIS LST (daytime) Dec
N01MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Jan
N02MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Feb
N03MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Mar
N04MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Apr
N05MSD3	MOD11A2	SD monthly MODIS LST (nighttime) May
N06MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Jun

Table 2.1: Covariates. (*continued*)

Name	Series	Attribute
N07MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Jul
N08MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Aug
N09MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Sep
N10MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Oct
N11MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Nov
N12MSD3	MOD11A2	SD monthly MODIS LST (nighttime) Dec
TMDMOD3	MOD11A2	Mean annual LST (daytime) MODIS
TMNMOD3	MOD11A2	Mean annual LST (nighttime) MODIS
P01CHE3	Global precipitation	Mean monthly precipitation at 1 km Jan
P02CHE3	Global precipitation	Mean monthly precipitation at 1 km Feb
P03CHE3	Global precipitation	Mean monthly precipitation at 1 km Mar
P04CHE3	Global precipitation	Mean monthly precipitation at 1 km Apr
P05CHE3	Global precipitation	Mean monthly precipitation

Chapter 3

Product specifications

3.1 Generic target specification

A global layer of harmonized national soil carbon stock maps has been developed according to the following specification:

- grid at 30 arc-seconds resolution (approximately 1 x 1 km). Generic grid has been provided by ISRIC World Soil Information;
- various SOC analysis methods and measurements are acceptable;
- 0–30 cm depth, including national increments and/or higher (deeper) depths where applicable;
- SOC stock [t/ha], bulk density (BD) [kg/m³] and stone content [percent] can be estimated or measured; and
- mapping/upscaling: various approaches possible (including country-specific stratification and custom resolution finer than 1 x 1 km).

More information about the product specification can be accessed at [FAO & GSP \(2017b\)](#).

3.2 Metadata specifications

In order to ensure that the national layers metadata would be sufficient for quality assessment and possible harmonization, the countries were required to share information about their original data according to the following principles:

1. Share auxiliary information about the national data sources, e.g. type of sampling (soil profile or auger), density of sampling points in the country, sampling design (distribution and sampling depth/s), time of sampling (year), selection criteria (if subset of soil profiles is selected from a larger national database).
2. Provide as much metadata as possible in order to estimate the quality of the global SOC map. For example, SOC method(s) of analysis.
3. Share metadata about SOC stocks calculation in terms of:
 - describing how SOC stocks for the target depth 0–30 cm have been calculated; if there are any deviations from this specification, provide an explanation;
 - quantifying the amount of carbon stored in litter (organic layer of forest floors);
 - if data allow, stratifying the national soil databases according to organic (peat) and inorganic soils, and estimate the SOC stock for peat soils to 1 m depth;
 - providing a description of the method used for bulk density measurements or estimations; and
 - providing a description of the method used for coarse fragments measurements or estimations.
4. Share details about the upscaling approach:
 - mapping method (description, citation); and
 - input data/covariates, grid, soil maps, etc.
5. In order to consider the temporal dimension of the SOC map, it is important to share the sampling date as metadata. If the national data situation allows, pre-1990 or post-1990 sub data sets might be defined. However, it

will be an important asset of this SOC map to demonstrate the density of existing soil carbon data sets. The more data points are used, the better the reliability and accuracy of the global product. Subsequent steps to improve the temporal dimension, will be considered at a later stage.

Chapter 4

Data collection and processing

4.1 Different scenarios of country driven action

The GSP Secretariat facilitated the process where countries were asked to deliver the following data and information:

- national soil organic carbon stock map;
- uncertainty assessment a) qualitative assessment (conventional mapping) and/or b) quantitative (digital soil mapping);
- metadata: The data shared by the counties are extensively documented to enable quality and uncertainty assessments. This will allow insights into the quality of the SOC maps, remaining gaps and harmonization needs. Countries were required to provide detailed metadata documented in the GSOCmap guidelines (FAO & GSP, 2017b); and
- one-page report: A brief report describes the current status of the national SOC data, data collection, preparation and harmonization efforts, selection of the method(s), challenges and assessment of the results.

The GSP Secretariat organized the data collection depending on national capacities, data availability/usability (Figure 4.1):

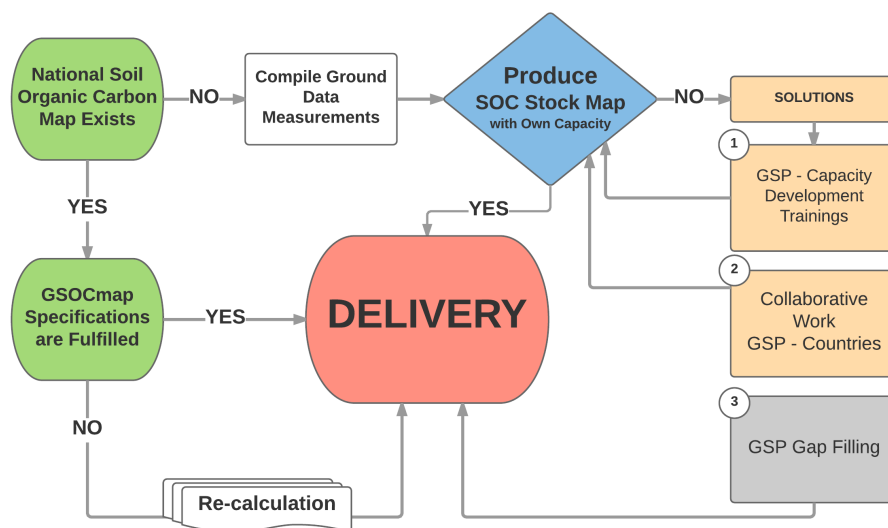


Figure 4.1: Country driven action depending on national capacity

- country submissions: Countries produced and delivered their GSOCmap contributions to the GSP Secretariat;
- joint efforts: The GSP Secretariat worked with the soil experts from the Members to produce their GSOCmap contributions; and
- GSP gap-filling: The Secretariat produced or used publicly available point or raster data for the countries that were not able to contribute to the current version of the product.

4.1.1 Delivery of the maps produced by the countries

The GSP Secretariat contacted countries about their potential contributions to the GSOCmap project and informed countries about the process and the procedure.

Countries already having national SOC maps that meet the specifications of this project, shared their data with the GSP Secretariat. If a national SOC map exists, and if not all requirements were met, adjustments to the existing SOC map were implemented (e.g. recalculation according to target depth).

Countries which did not yet have a national SOC map, developed such a map based on the specifications. Where needed, the GSP Secretariat supported such national activities by organising training sessions.

Upon receiving the national maps, the GSP Secretariat undertook the preliminary data quality checks. Whenever the national maps were inconsistent with the GSOC specifications, the GSP Secretariat worked in close collaboration with the institutions that provided the national data to resolve the existing issues.

76 countries submitted their maps as a contribution to the GSOCmap. This represents 65 percent of the world area (Table 4.1).

4.1.1.1 Data submission form

To deliver their data, countries were required to use the online data submission tool. The tool realized a guided delivery process which required the submitter to upload the map along with a one-page report, and to answer questions about the methodology according to the metadata specifications. The questionnaire can be found in Annex A.

This questionnaire and the report were used to create the country-specific metadata for the map. A summary of this data can be found in Annex D and chapter 5.

4.1.2 Joint efforts

If the in-country development of a SOC map was not possible due to insufficient capacity, the original SOC measurements were shared with the GSP secretariat which would then execute the mapping in close cooperation with the national GSP-focal points and/or institutional data providers.

For the mapping, the GSP Secretariat used state-of-the-art digital soil mapping techniques and publicly available layers of environmental covariates (ISRIC, ftp service). The data were then evaluated by the national experts who made the final decision regarding the results of the joint mapping procedure and the final submission of the map. 8 country maps were prepared with joint efforts between the GSP and the countries (Table 4.1).

4.1.3 GSP gapfilling

For the countries that could not provide a SOC map or any original measurements, the GSP Secretariat used one of the two gap-filling approaches: spatial modeling using publicly available data or, in the case of absence or insufficient amount of data, using publicly available SOC stock maps.

The gap filling procedure involved producing maps for 108 countries. 69 country maps (26.4 percent of the world area) were done using available data; and 39 country maps (1.0 percent of the world area) were filled using external datasets (e.g. soilgrids.org data).

Table 4.1: Sources of the country maps included in the GSOCmap v1.6

Contribution	n	Area sqkm	Percent of world area
Country submission	74	83 821 626	64.8
Joint effort with GSP	13	10 029 039	7.8
GSP gap-filling	69	34 183 499	26.4
External dataset: soilgrids.org	39	1 304 964	1.0

4.1.3.1 Spatial modeling using publicly available data

For the countries where publicly available data of SOC measurements were sufficient for SOC mapping, the GSP Secretariat used digital soil mapping techniques to create the maps. The following data sources used for this purpose:

- WOSIS (Batjes *et al.*, 2017);
- LUCAS soil (European Union – EU27) (Toth *et al.*, 2013); and
- AfSIS (Africa Soil Information Service) (Walsh *et al.*, 2009).

Publicly available layers of environmental covariates ([WorldGrids.org](https://worldgrids.org) currently available at <https://zenodo.org/record/1637816#.XP4tXogzbiU> and <https://gitlab.com/openlandmap>) were used for the spatial modeling. Example scripts similar to the ones used by the GSP for building these maps can be found in Annex C.

4.1.3.2 Using publicly available SOC stock maps

For the countries lacking any publicly available data of SOC measurements (around 1.4 percent of the global coverage), the maps were produced using SoilGrids 250m product (Poggio *et al.*, 2021) resampled according to the GSOC specifications.

4.2 Data processing and compilation of the GSOCmap

In order to compile a Global Soil Organic Carbon Map from the national contributions, the GSP Secretariat conducted basic data processing aimed to standardize the individual layers with minimal or no changes to the data provided by the countries. The processing steps included:

- reprojecting data to coordinate system lon/lat WGS84 with spatial resolution of 30 arc-second using ‘nearest neighbour’ method (where necessary);
- resampling the data to 30 arc seconds grid resolution using bilinear interpolation (in case the cell size of the original data was different);
- converting SOC stock values to tonnes/ha (where necessary);
- mosaicking individual maps of the countries to acquire the global layer using ‘nearest neighbour’ method for resampling;
- filling NoData values at national borderlines (in case the countries didn’t use suggested empty grids) using GDAL gapfilling algorithm which interpolates values for all designated NoData pixels using inverse distance weighting and a four direction conic search to find values to interpolate from. A mask compiled from a 5-km buffer around country borders (excluding water bodies and coastlines) was used as a gap-filling procedure in order to make sure that only border gaps are filled and the NoData values provided by countries because of lack of information are preserved; and
- applying a global mask of water bodies (World Water Bodies, Esri, Garmin International).

Chapter 5

Metadata

The GSOCmap is a compilation of soil organic carbon stock maps produced by the countries in accordance with the GSOCmap guidelines (FAO & GSP, 2017b). The total number of profiles/sampling locations used to create the global product is: 1 079 617. The number of sampling plots was calculated from the meta-data provided by the Members.

5.1 Sampling density

The metadata allows to assess the density of sampling points per country as shown in Figure 5.1. It varies greatly and reflects the differences in the soil data coverage between countries and regions. This information can be used as an assessment of the current status of the available soil information in the world and to identify the regions where additional sampling is most needed.

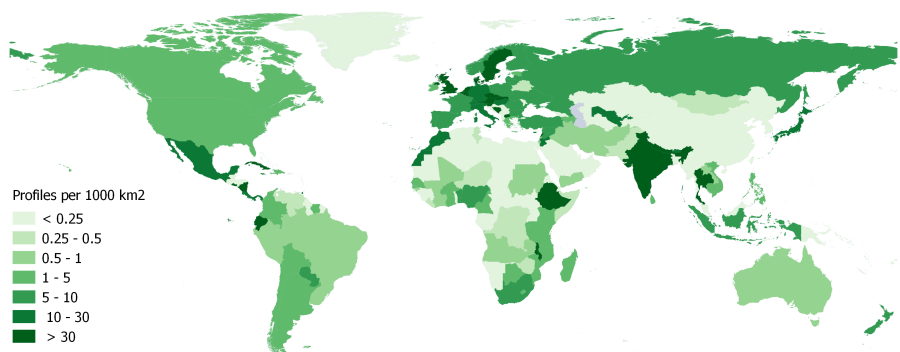


Figure 5.1: Density of point data (per country). Source: UN Geospatial, 2020, modified by authors.

However, aggregation of the data at the country level does not allow to accurately represent the sampling density in case of uneven distribution.

5.2 Temporal dimension

The metadata on the period of sampling was analyzed against the suggested baseline date 1990. 11 percent of the countries did not provide information about the temporal dimension. Besides, data about the temporal dimension for the 25 percent of the countries with data from external datasets, like soilgrids.org, were not available. The other 65 percent is divided: 15 percent with data from before 1990; 25 percent with a mix of data before and after 1990 and only 25 percent with all the data surveyed after 1990.

The results show that most countries had to include observations from before 1990 to develop a dataset representing all their territory (Figure 5.2). This means that the GSOCmap can be used as a baseline for SOC monitoring only for the countries with all the submitted data originating from recent soil surveys. However, the GSOCmap can be viewed as a baseline map, as it contains the best available estimation of SOC at the country level (see Validation and comparison with existing products), making it an important tool for identifying SOC deficient areas within the countries and subsequently for planning the soil-protecting, sampling and monitoring activities.

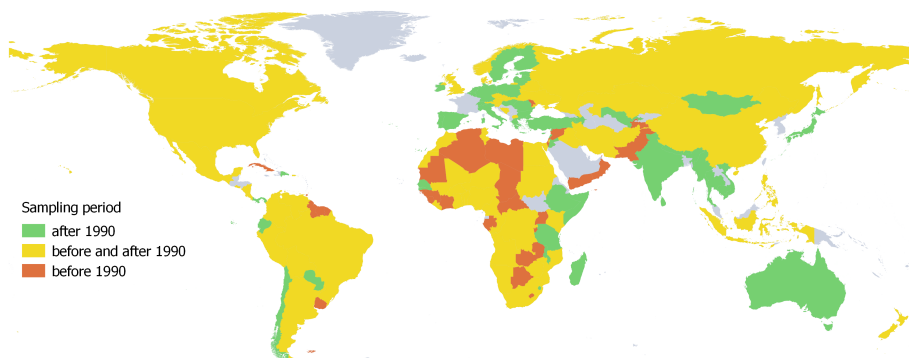


Figure 5.2: Sampling period. Source: UN Geospatial, 2020, modified by authors.

5.3 Soil organic carbon

Concerning the SOC analysis method, 42 percent of the countries used wet oxidation, and 14 percent used dry combustion. The number of countries using soil spectroscopy is less than 1 percent.

The metadata shows differences in the methods used throughout the world for determining SOC (Figure 5.3). This is valuable information for investigating the possibility of further harmonization of national data. One of the known issues is the difference in measured carbon values between Ethiopia and the surrounding countries, which could be caused by the difference in SOC analysis method since Ethiopia was the only country to use soil spectroscopy measurements as its primary data source.

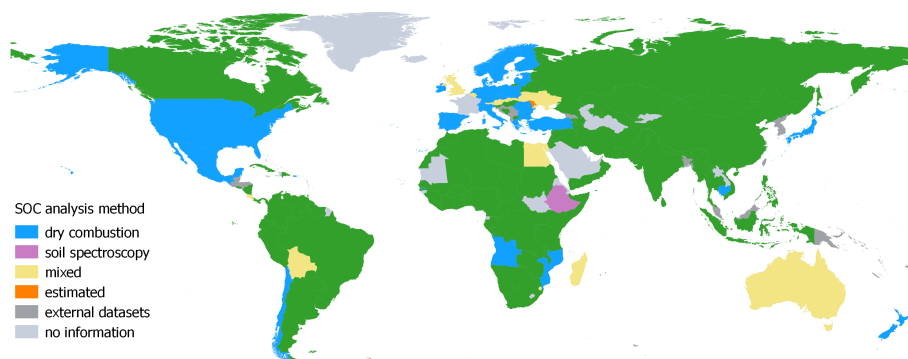


Figure 5.3: SOC analysis methods. Source: UN Geospatial, 2020, modified by authors.

5.4 Bulk density

Measurements of bulk density were not available in many countries, thus different strategies were used to overcome this limitation (Figure 5.4).

1. Using nationally developed pedotransfer functions.
2. Using pedotransfer functions suggested in the cookbook manual.
3. Using the values from publicly available datasets (such as Harmonized World Soil Database (FAO, 2012) and SoilGrids (Hengl *et al.*, 2017)).

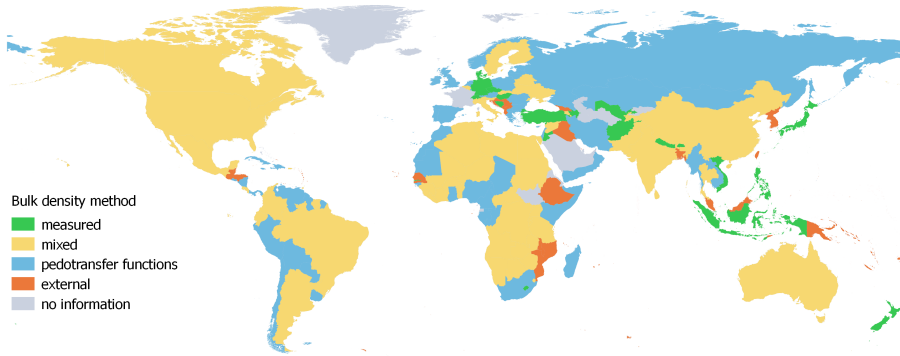


Figure 5.4: Bulk density analysis methods. Source: UN Geospatial, 2020, modified by authors.

Only 8 percent of the countries used only measured bulk density data to estimate the soil organic carbon stock. 27 percent submitted measured values for some profiles, but had to use pedotransfer functions for others. 28 percent relied only on pedotransfer functions. And external datasets like soilgrids.org or the HWSD were used for 28 percent of the countries. 9 percent of the countries did not provide information about the source of their bulk density data.

The estimation of bulk density is a potential source of high uncertainty in the calculation of carbon stocks, especially in soils with high stoniness (Poeplau *et al.*, 2017). According to our findings, more than 55 percent of the countries used pedotranfer functions, but only 25 percent used locally fitted pedotranfer functions. With a high percentage of countries using the pedotransfer functions suggested in Yigini *et al.* (2018).

5.5 Coarse fragments

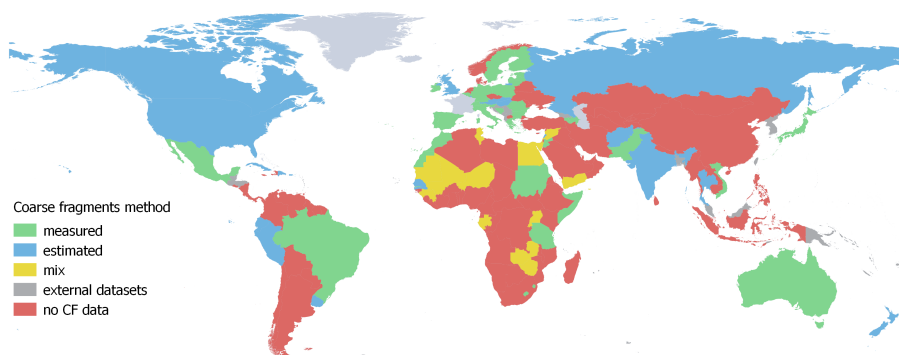


Figure 5.5: Coarse fragments methods. Source: UN Geospatial, 2020, modified by authors.

The metadata show that the majority of the countries had limited data on the coarse fragment content which could be a source of uncertainty in the calculation of organic carbon stocks, especially in mountainous areas (Figure 5.5). Only 17 percent of the countries had measured data for the amount of coarse fragments. 10 percent used estimated values and 7 percent used a mix between estimated and measured values. Almost 40 percent of the countries did not use any information about the coarse fragments fraction for the organic carbon stock estimation.

5.6 Mapping methods

Various mapping methods were used by the countries depending on the capacity, data availability, and the specific features of the local soil cover, including but not limited to:

- conventional upscaling: geo-matching, class-matching;

- digital soil mapping methods: Multiple linear regression, regression kriging, multivariate adaptive regression splines, generalized linear models, generalized additive models, etc.;
- machine learning techniques: Random forest, support vector machine, neural networks, regression trees, bayesian trees, etc.;
- ensemble models combining different DSM methods; and
- geostatistical methods: Ordinary kriging, IDW.

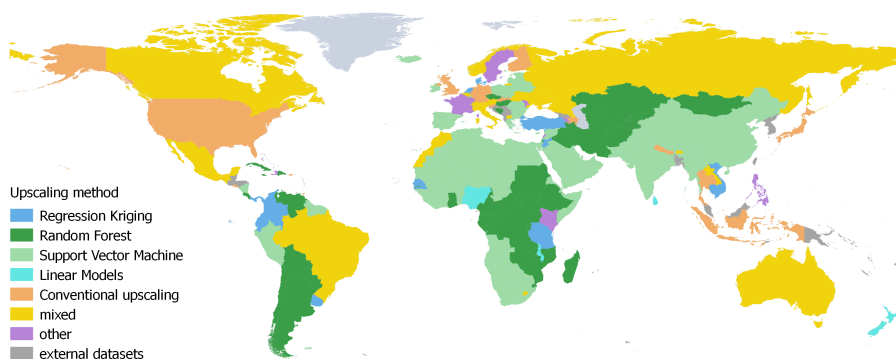


Figure 5.6: Mapping methods. Source: UN Geospatial, 2020, modified by authors.

Figure 5.6 shows that most countries (66 percent) were able to use the state-of-the-art digital soil mapping techniques which demonstrates the overall success of the capacity building program undertaken by the FAO/GSP. Only 7 percent of the countries used conventional upscaling.

The heterogeneity of the mapping techniques could be one of the sources of uncertainty and ‘border effects’ between national products. However, the map shows that the difference in mapping methods is not the primary source of border inconsistency. As shown in the Figure 5.7, in many cases, the maps produced with different mapping methods have comparable values and form a continuous surface of organic carbon distribution with acceptable differences. Besides, there is no best mapping method for digital soil mapping, and testing and selection has to be done for every data scenario (FAO & GSP, 2017b).

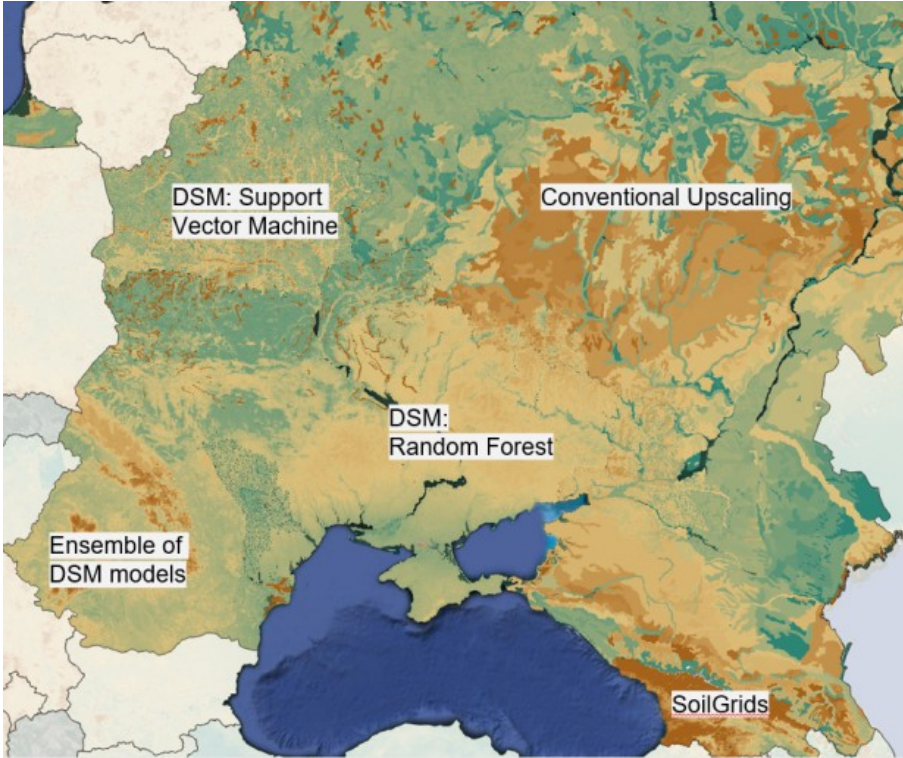


Figure 5.7: Fragment of the GSOCmap with different techniques used for mapping at country level

The primary source of uncertainty and border inconsistencies appears to be in the original point data quality and representativity. The difference at the borders between the countries occurs when the adjacent region is not covered with soil sampling data and the values are extrapolated using a model from a different area or assigned on the basis of expert knowledge. Therefore, it is suggested that the work on improving the global consistency should be primarily focused on acquiring additional data in the under sampled regions and capacity development aimed at ensuring that an appropriate mapping method was used based on the data distribution and representativity.

Chapter 6

Results

6.1 Total global soil organic carbon stock

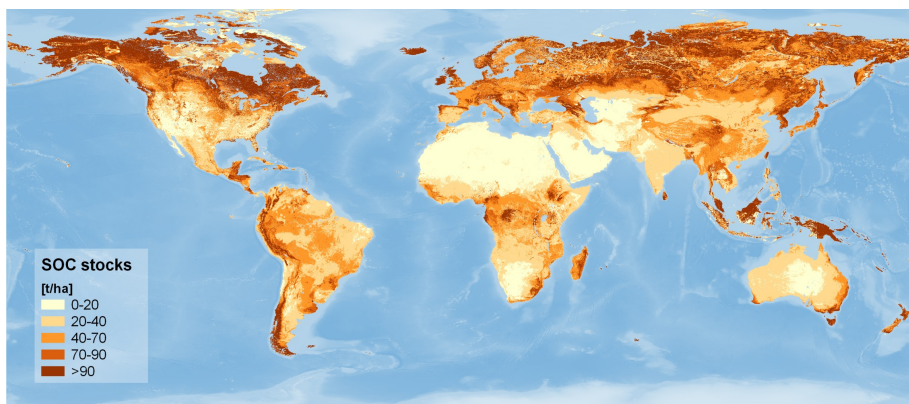


Figure 6.1: GSOCmap version 1.6.0.

Global soil organic carbon stock for topsoil (0 to 30 cm) is **682** Petagrams. This value is 2.4 percent lower than the value for the HWSDa (Köchy *et al.*,

2015) (Table 6.1).

Table 6.1: Global SOC estimates.

GSOCmap	HWSD	HWSDa	FAO2007	WISE	DSMW	soilgrids 250m
682	967	699	710	504	574	1 267

6.1.1 Statistics for countries (GSOCmap V 1.6.0)

Statistics for the countries were calculated based on the global administrative units layer as the source for country boundaries. Over 70 percent of the global SOC stocks at 30 cm is held by 14 countries: the Russian Federation, Canada, the United States of America, China, Brazil, Indonesia, Australia, Argentina, the Democratic Republic of the Congo, Kazakhstan, Peru, Mongolia, Chile, and India (Table 6.2). Among these countries, Indonesia and Chile have the highest mean SOC stocks (103.6 and 71.8 t/ha respectively).

Table 6.2: SOC stocks for selected countries.

	SOC stock (Pg)	Mean SOC (t/ha)	Global share (%)
Russian Federation	147.0	36.7	21.6
Canada	80.2	37.9	11.8
United States of America	54.0	34.9	7.9
China	44.8	33.2	6.6
Brazil	35.1	34.9	5.2
Indonesia	22.5	103.6	3.3
Australia	22.5	22.6	3.3
Argentina	13.7	34.7	2.0
Democratic Republic of the Congo	12.3	45.6	1.8
Kazakhstan	11.9	25.7	1.7
Peru	10.0	65.9	1.5
Mongolia	8.1	30.8	1.2
Chile	7.8	71.8	1.1
India	7.5	20.1	1.1

6.1.2 Statistics for climate zones, land use and soil types

To estimate the relationship between SOC and climate, the result layer was spatially intersected with the IPCC, 2006 climate regions layer. The zones are defined by a set of rules based on annual mean daily temperature, total annual precipitation, total annual potential evapotranspiration (PET) and elevation. The largest carbon pool is located in the boreal moist zone followed by the Cool Temperate Moist zone (Table 6.3). To show the relation between below ground and above ground organic carbon stocks, SOC stocks per IPCC climate region were compared to aboveground organic carbon stocks derived from IPCC (2008) (Figure 6.3).

Table 6.3: SOC stocks by IPCC climate region.

	SOC stock (Pg)	Mean SOC (t/ha)	Global share (%)
Boreal Moist	143.0	40.1	21.0
Cool Temperate Moist	97.7	45.1	14.3
Tropical Moist	86.3	46.3	12.7
Subtropical Moist	56.5	45.6	8.3
Cool Temperate Dry	51.3	28.9	7.5
Warm Temperate Moist	42.4	42.1	6.2
Boreal Dry	40.0	38.0	5.9
Subtropical Dry	36.7	21.7	5.4
Tropical Dry	32.6	19.9	4.8
Warm Temperate Dry	31.3	22.1	4.6
Polar Moist	28.7	29.2	4.2
Polar Dry	13.5	32.6	2.0

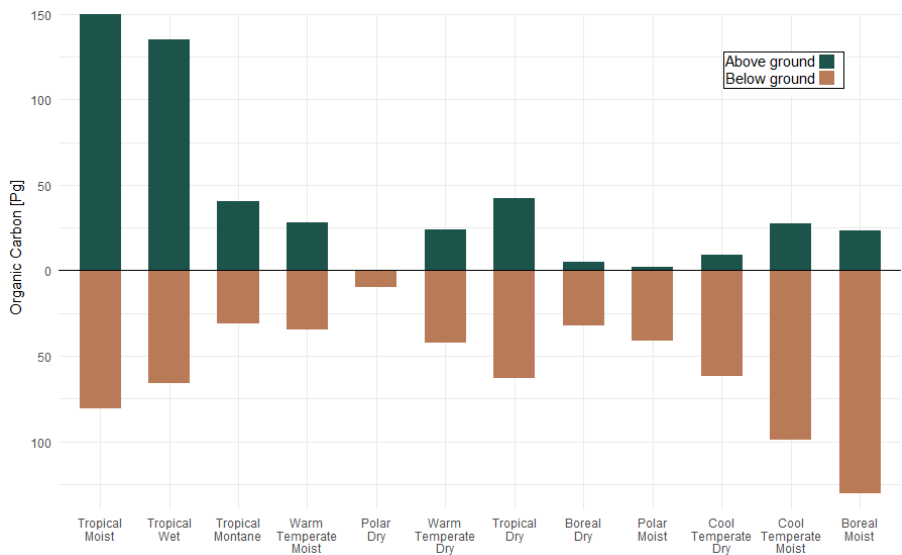


Figure 6.2: Soil organic carbon stocks per IPCC climate region calculated from the GSOCmap v1.6.0 and aboveground organic carbon from IPCC (2008)

The same approach was used to compare the GSOCmap to land cover data. The land covers were based on is the reclassified GlobCover 2009 map (Arino *et al.* 2009). The following land cover classes were considered: cropland, forest, grassland, wetland and other and reclassified from the from the Globcover classes (Table 6.4). The greatest share of SOC is found within forests (298.8 Pg, 43.1 percent) followed by grasslands (179.4 Pg, 25.9 percent) and croplands (143.4 Pg, 20.7 percent). The greatest mean SOC stocks in t/ha are found in wetlands with 95.4 t/ha.

Table 6.4: SOC stocks by land cover.

	SOC stock (Pg)	Mean SOC (t/ha)	Global share (%)
Forests	291.6	42.1	42.8
Grasslands	178.2	30.1	26.1
Croplands	140.1	35.7	20.5
Others	46.7	14.8	6.9
Wetlands	24.0	47.8	3.5

The relationship with soil types was explored using the Harmonized World Soil Database (HWSD) (FAO/IIASA/ISRIC/ISSCAS/JRC, 2012). The HWSD is a 30 arc-second raster database with over 16 000 different soil mapping units that combines existing regional and national updates of soil information world-wide (SOTER, ESD, Soil Map of China, WISE) with the information contained within the 1:5 000 000 scale FAO-UNESCO Soil Map of the World (FAO, 1971–1981). Although the soils richest in organic carbon are Histosols, Gleysols, Andosols and Chernozems, most of the carbon in the world is stored in Leptosols and Cambisols due to their larger area coverage (Table 6.5).

Table 6.5: SOC stocks by soil type.

	SOC stock (Pg)	Mean SOC (t/ha)	Global share (%)
Leptosols	78.7	38.8	11.5
Cambisols	65.8	46.0	9.7
Gleysols	60.3	41.0	8.8
Podzols	56.0	45.6	8.2
Acrisols	45.2	26.9	6.6
Regosols	44.5	40.9	6.5
Ferralsols	39.8	30.7	5.8
Luvisols	36.2	63.8	5.3
Histosols	34.6	19.4	5.1
Arenosols	24.1	15.5	3.5
Calcisols	21.3	49.1	3.1
Chernozems	19.6	34.1	2.9
Fluvisols	19.4	41.2	2.8
Phaeozems	18.3	28.6	2.7
Kastanozems	16.7	31.0	2.5
Podzoluvisols	15.9	25.3	2.3
Vertisols	9.6	31.3	1.4
Lixisols	9.3	27.9	1.4
Solonetz	8.3	56.2	1.2
Andosols	8.2	44.0	1.2
Nitisols	7.7	36.9	1.1
Plinthosols	6.0	10.9	0.9
Greyzems	5.2	36.1	0.8
Planosols	4.3	34.5	0.6
Gypsisols	3.1	33.0	0.5
Solonchaks	2.6	42.7	0.4
Alisols	2.4	28.8	0.4
Anthrosols	2.2	22.9	0.3

6.2 Validation, uncertainty and comparison with existing products

6.2.1 Validation of the GSOCmap using available data

The spatial distribution of soil organic carbon (SOC) represents one of the largest uncertainties in the carbon cycle (Figure 6.4). High resolution gridded datasets of SOC are increasingly important for global modeling efforts and validation strategies (Jackson *et al.*, 2017). Validation experiments (e.g. across borders), the comparison of different approaches to predict soil carbon and the continuous calibration of country-specific to regional to global models are required to provide reliable estimates and enable the monitoring of SOC stocks.

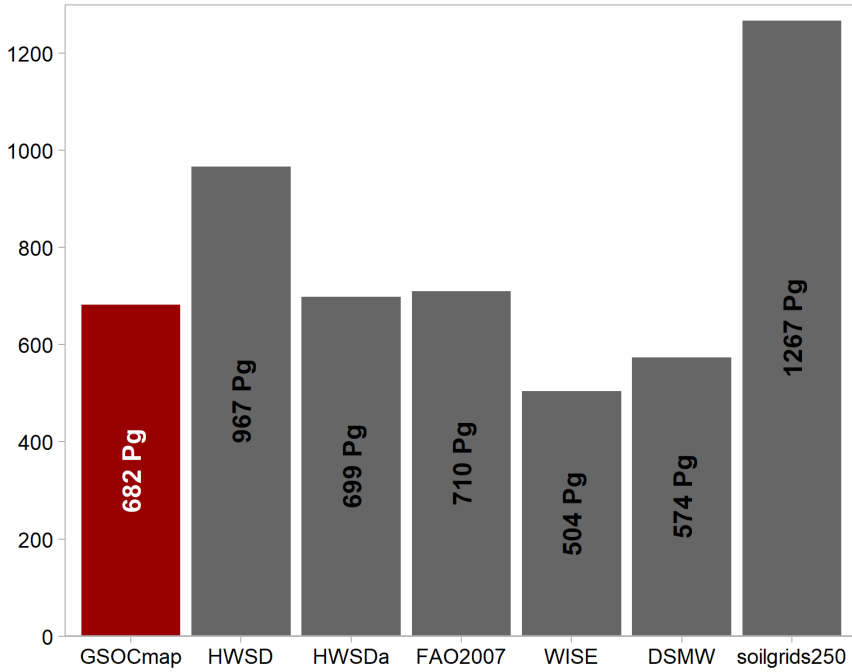


Figure 6.3: Comparison between different estimates of global SOC stocks in topsoils

To compare and test different approaches (e.g. modeling and geo-matching), to map SOC stocks is relevant to reduce the current levels of uncertainty regarding the spatial variability and distribution of SOC, because they will work differently for the same objective. Using the same dataset, different approaches to map SOC will share bias derived from the quality of the data and the data characteristics that allow to meet modeling assumptions, or provide certainty to the soil mapper delineating a soil carbon polygon unit.

The validation of the map was done by comparison with available soil profile data and the existing global SOC products. First, a validation dataset was compiled from international publicly available datasets, in particular the International Soil Carbon Network (ISCN) (Nave *et al.*, 2017; Boby *et al.*, 2010; Bockheim *et al.*, 2003; Bockheim *et al.*, 2010; Buell *et al.*, 2004; Camill

et al., 2009; Cole *et al.*, 2013; Harden *et al.*, 1999; Harden *et al.*, 2012; Johnson *et al.*, 2011; Jorgenson *et al.*, 2009; Kane *et al.*, 2009; Manies & Survey, 2004; Myers-Smith *et al.*, 2007; O'Donnell *et al.*, 2011; Ping & Liang, 2011; Tarnocai *et al.*, 2009; Trumbore *et al.*, 1999; Zinke *et al.*, 1986) including extensive data contributions to the ISCN from the USDA Natural Resources Conservation Service, National Cooperative Soil Survey (Soil Survey Staff, 2014); World Soil Information Service (WoSIS) (Batjes *et al.*, 2017), soil organic carbon stock estimates with uncertainty estimation across Latin America (Guevara *et al.*, 2019), Latin America and the Caribbean's Soil Information System (SIS-LAC)(GSP, 2013), the Northern Circumpolar Soil Carbon Database (Hugelius *et al.*, 2013) and Soil Profile Analytical Database of Europe of Measured parameters (SPADE/M) (Hiederer *et al.*, 2006). From these databases only the profiles with measured SOC and bulk density values were selected, since these parameters are essential for calculating SOC stocks – 50 708 points in total (Figure 6.5). Coarse fragments data were used in SOC stock calculations, where available.

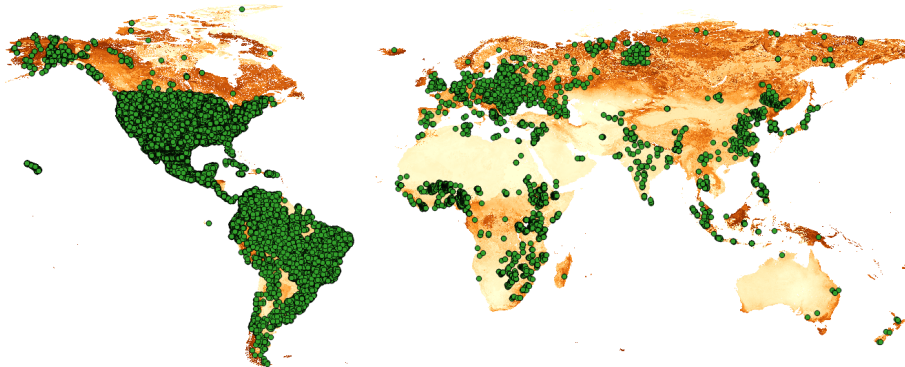


Figure 6.4: Locations of points in the validation dataset based on publicly available data

Mass preserving spline functions were applied to estimate the carbon content, bulk density and coarse fragments of a standardized horizon of 0 to 30 cm. Finally, the organic carbon stock was estimated using the GSIF R package. These values were compared with the values in the GSOCmap. The same

dataset was used to compare validation results with 2 other global recent products of SOC, one derived from the Harmonized World Soil Database (HWSD), based on soil type polygon units (Köchy *et al.*, 2015) and a second product derived from the SoilGrids initiative based on machine learning and environmental correlation (Hengl *et al.*, 2017; Poggio *et al.*, 2021). The three SOC maps were resampled from their original resolution to a 1 x 1 km grid and centered on the same spatial extent. This comparison allows to estimate the accuracy of the GSOCmap in relation to other global products and to ground measurements. The results of the validation analysis are shown in table (Table 6.6).

Table 6.6: Comparison of validation criteria between the three SOC products based on international publicly available datasets: FAC2 – fraction of predictions within a factor of two; MB – mean bias; MGE – mean gross error; NMB – normalized mean bias; NMGE – normalized mean gross error; RMSE – root mean squared error; IOA – index of agreement (Willmott *et al.*, 2012).

Map	FAC2	MB	MGE	NMB	NMGE	RMSE	IOA
GSOCmap	0.62	-0.39	35.07	-0.01	0.66	60.71	0.50
SoilGrids	0.47	31.12	48.76	0.59	0.92	67.88	0.30
HWSD	0.57	12.22	44.01	0.23	0.83	75.42	0.37

The result show that GSOCmap on average has lower error than other global products. Mean gross error (mean absolute error) for GSOCmap is around 35 t/ha, while other global products have error more than 40 t/ha. Root mean squared error is significantly higher than mean absolute error for all products which indicates the presence of a relatively small number of very high errors, most likely associated with organic soils. Global modeling approach of SoilGrids results in the highest mean gross error, while polygon-derived HWSD has higher root mean squared error. However, GSOCmap shows lowest errors in both case which suggests that country-driven approach allows to minimize errors at the global scale. The FAC2 (that takes a value of 1 in a perfect model) indicates the fraction of predictions within a factor of 2 of the observed value. The FAC2 shows a better performance of the GSOCmap. Another important observation is that GSOCmap is almost unbiased (mean bias -0.39 t/ha) while HWSD and especially SoilGrids have significant positive bias (12.21 t/ha and 31.12 t/ha respectively).

To assess the overall quality of the model, Index of Agreement based on Willmott *et al.* (2012) was applied to all three global products. This index between -1 and +1 with values approaching +1 representing better model performance. GSOCmap showed the highest IOA performance among the three analyzed products with the score 0.5, which indicates that the sum of the error magnitudes is one half of the sum of the observed deviation magnitudes.

These results show that the GSOCmap country-driven approach allows to use much more data locally which helps to minimize errors and ensure better representation of the map. However, since the publicly available datasets mostly cover North and South America and have large gaps in African and Eurasia, the validation dataset may be biased. Therefore, GSP started collecting national datasets that were not used for the development of the GSOCmap, for validation.

For the global comparisons, the analysis based on numerical confusion matrices derived at the pixel level on SAGA GIS (Conrad *et al.*, 2015). A confusion matrix is a specific table layout that allows visualization of the main changes from a reference map (initial state) to another (final state). Thus, it quantifies the absolute difference and the direction of change comparing the GSOCmap with the HWSD, the GSOCmap with SoilGrids and the HWSD with SoilGrids. A map of (positive and negative) changes is derived for each iteration (between the reference and the final state), where values close to 0 represent areas of high agreement between the two compared products.

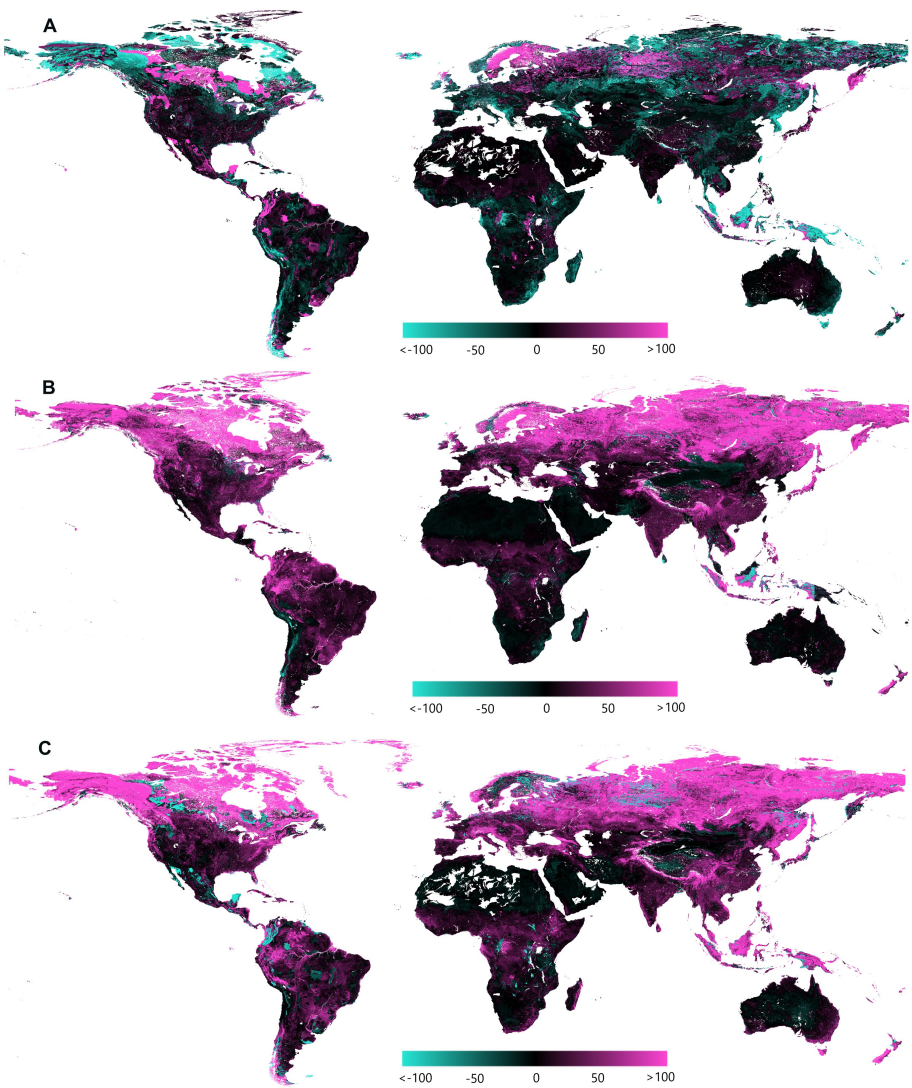


Figure 6.5: Change vector maps. Derived from a standardized confusion matrix. The map in A shows the changes from GSOCmap to the HWSD, B from the GSOCmap to SoilGrids and C from HWSD to SoilGrids.

It was found that there is a larger agreement between the GSOCmap and HWSO than between the GSOCmap and SoilGrids (Figure 6.6A). While positive and negative changes from the GSOCmap to the HWSO are irregularly distributed, the changes from the GSOCmap to the SoilGrids products tend to be positive, suggesting a major carbon pool predicted by the machine learning approach (Figure 6.6B). A similar pattern was found by analyzing the changes from HWSO to SoilGrids map (Figure 6.6C).

These results are useful to identify and quantify differences among products that can be used to inform the development of future versions of country-specific and global SOC mapping efforts.

6.2.2 Uncertainty analysis

Uncertainty analysis aims at quantifying possible deviations of SOC stock estimation on the maps from the real values. SOC map uncertainties come from the soil sampling, measurements of soil properties, and mapping techniques.

Although the GSP Secretariat asked the countries to provide information about the uncertainty of their contributed maps, not many countries were able to provide it. Besides, the layers contributed from the countries which could deliver, were generated using different and not comparable methodologies. The different methodologies used included: confidence intervals for the SOC values; standard deviation from regression kriging; standard deviation from an ensemble of different DSM models, and uncertainties as percentage based on expert knowledge. Therefore, a global approach for estimating uncertainties was required.

The uncertainty of the map was estimated by spatial comparison with all available SOC data. For this purpose, several available global, regional and national soil databases were merged, including WoSIS soil profile database (Batjes *et al.*, 2017), LUCAS Soil Database (Toth *et al.*, 2013), Soil Information System for Latin American and the Caribbean (SISLAC) (GSP, 2013), Northern Circumpolar Soil Carbon Database (Hugelius *et al.*, 2013), as well as national datasets provided by participating countries. This database of 276 028 points was used to calculate the residuals of the maps. The residuals were estimated as the difference between the measured and predicted values and expressed in percent of the predicted values by the GSOCmap. The current map presented in Figure 6.7 is the result of the interpolation using ordinary kriging of the residuals. The map shows standard deviations in tonnes/ha.

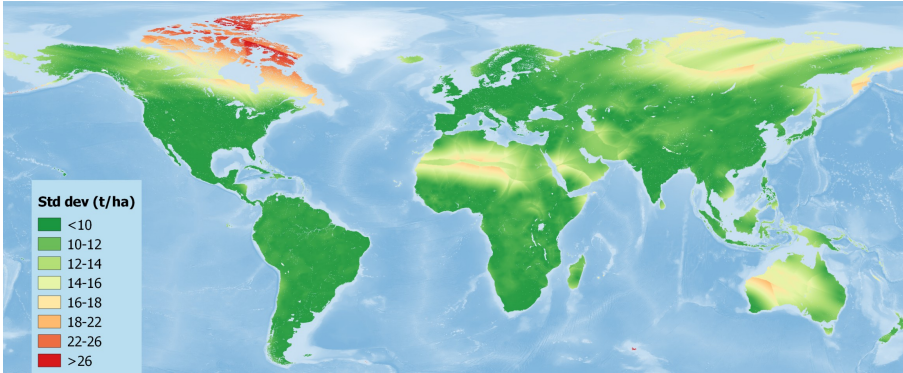


Figure 6.6: Map of standard deviations

The current assessment shows highest uncertainty values in the tropical desert and arctic desert areas, due to insufficient number of soil samples from these regions. The difference in uncertainties between the countries in non-desert areas is mainly associated with the density of soil sampling and with the choice of mapping techniques. Based on the map, the uncertainty of global SOC pool value is estimated as ± 144 Pg or around 20 percent.

The uncertainty layers provided by 22 countries (Argentina, Australia, Austria, Belgium, Brazil, Canada, Colombia, Denmark, Ecuador, France, Germany, Hungary, India, Indonesia, Italy, Mexico, Netherlands, Nicaragua, Panama, Russia, Venezuela, Viet Nam) can be requested from the data listed providers in Appendix D.

As part of the envisioned future updates, which will make up the GSOCmap v2.0, a harmonized approach will be implemented to support countries in producing the mandatory uncertainty layers of their national SOC stock maps.

6.3 GSOCmap web services

The GSOCmap Web Services portal is operational with basic functionality while GLOSIS portal is under development (Figure 6.7). The map is available online with functionality allowing to view, query and download the data. It provides in-

formation about data sources acknowledging all organizations which contributed to the map (Figure 6.8).

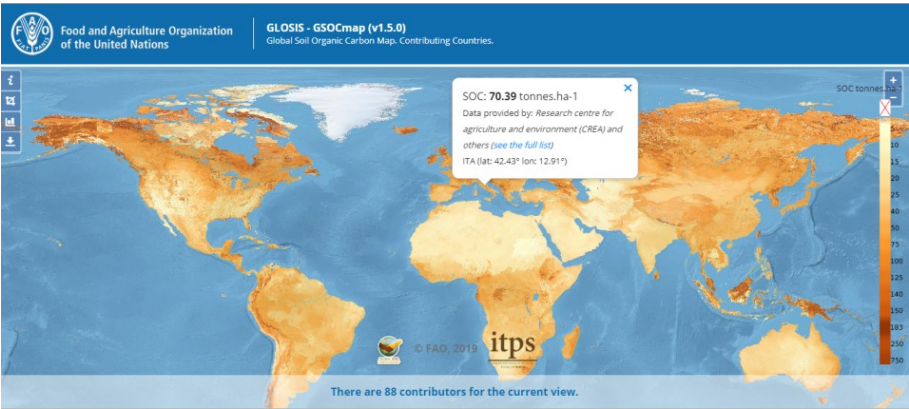


Figure 6.7: Accessing the GSOCmap on the web

GSOCMAP WEB SERVICES ARCHITECTURE

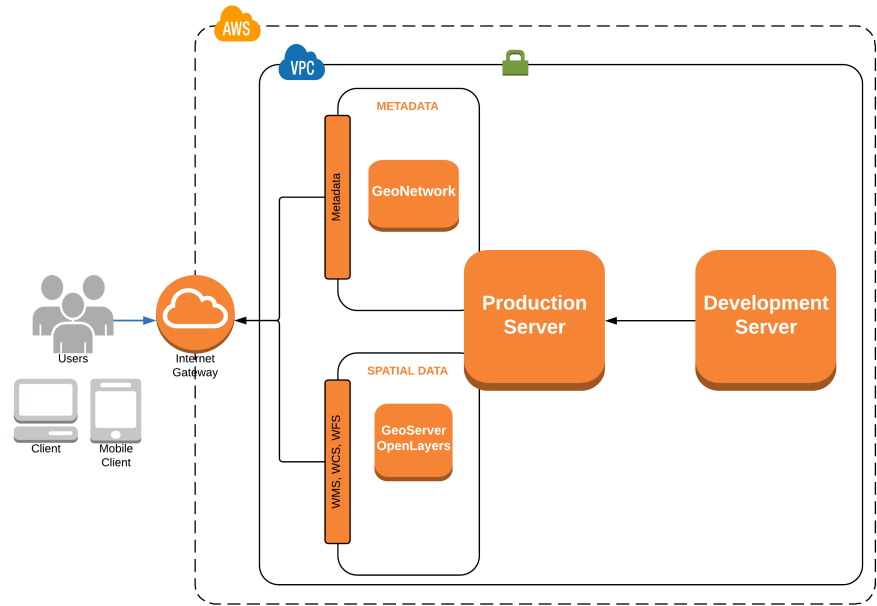


Figure 6.8: GSOCmap web services architecture

Chapter 7

Conclusions and the way forward

The GSOCmap is a product of global efforts made to bring together the existing knowledge about soil organic carbon from all over the world. The map is currently based on more than 1 million profiles, most of the area is covered by original maps produced by the countries. This ensures that the GSOCmap is a global product which is consistent with the national soil knowledge and gives the best available estimation of SOC stocks at the country level.

The known issues are the differences at the borders between certain countries and overestimation of SOC stocks for the countries where external datasets were used for gap-filling. These issues will be gradually addressed as more data is collected by countries which will allow to improve national maps and replace gap-filling with original data. A harmonized approach will be implemented to guarantee and support the production of accompanying and mandatory uncertainty estimations of the national SOC stock maps as part of the upcoming GSOCmap 2.0.

The GSOCmap is to be continuously improved as the countries gather more data to improve their maps. The versioning system is being implemented which implies publishing the latest version of the map and keeping all previous versions available upon request.

Versioning system

The GSP uses semantic versioning at certain level so that there is a standard pattern to data releases. Semantic versioning is widely used in the software development world and helps developers having a standardized way of versioning software releases. It follows the format of MAJOR.MINOR.PATCH.

The product released on World Soil Day 2017,(5 December) as V1.0.0 and was soon updated to V1.2.0 with minor updates. In June 2019 the map was updated to V1.5.0 with new and updated national contributions, as well as improved gap-filling. The current update to v1.6 presents a further improvement and update of the GSOCmap regarding national submissions as well as gap-filling and external datasets. Change-log is also to be released along with the data with each public release. The current changelog is presented in Annex 10. The future releases will be using the following logic:

Major Major version will be incremented with substantial updates.

Minor Minor version will be incremented with new country submissions, replacements

Patch Patch version will be incremented with error fixes (i.e. removing outliers, fixing calculation errors, etc.)

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Annex A: Metadata questionnaire

A1: Source data

- Total number of soil profiles/sampling locations;
- Type of sampling (profiles/augers/topsoil);
- Number of locations for each sampling method;
- Sampling Period (e.g. 1980–2007);
- Georeferencing (GPS coordinates/Location names);
- Depth of sampling;
- Sampling design (e.g. transect, catena, land use etc.)

A2: Analysis methods

- Methods of Soil Organic Carbon analysis;
- Methods of Bulk Density analysis (measured/estimated)
 - Details about the sampling;
 - Pedotransfer functions, default values, citations;

- External data-sets (HWSD, SoilGrids.org);
- Methods of Coarse Fragments (measured/estimated/NA)
 - Coarse fragments unit (e.g. % volume / % weight)
- Peat (sampling and description method);

A3: Mapping

- mapping method (DSM / Conventional upscaling)
 - The method(s) used (e.g. Multiple linear regression, Regression-Kriging, Random Forest...);
- Map quality measures (Digital Soil Mapping)
 - Mean error (ME), Mean absolute error , root mean squared error, amount of variance explained;
- Units (tonnes/ha, kg/m²);
- Resampling Method (if used)

A4: Contact details

- Submitter contact details;
- Institute (Data Holder / Handler);
- Citation;
- Update Frequency;
- Comments, Remarks

Annex B: Changelog file of the GSOCmap

GSOCmap:

change log. From 30/05/2019. Most recent changes first / on top.

VERSION v1.6.0

New submissions: KOR, GEO

Updated national maps: URY, TUR, MKD, LAO, VNM

Updated gap-filling: GTM, BLZ, HND (regional DSM); PRK, TWN, DZA, TUN, LBY, EGY, ISR, MRT, MLI, BFA, NER, TCD, ERI (regional DSM + global CLM5)

Updated external datasets (replaced Soilgrids with a new version):

MYS, BGD, PNG, CCK, CXR, PLW, SP-, PIS, ABW, AIA, ATG, BES, BLM, BMU, BRB, CUW, CYM, DMA, GRD, KNA, LCA, MAF, MSR, MTQ, SXM, TCA, VCT, VGB, ATF, COM, HMD, IOT, MUS, MYT, REU, SYC, COK, FJI, FSM, KIR, MHL, NCL, NFK, NIU, NRU, PCN, PYF, SLB, TKL, TON, UMI, VUT, WLF, WSM, XCL, STP, CPV, BVT, SGS, SHN, SPM, GLP, SRB, XKO, MNE, HRV, ALB

VERSION 1.5.0

* New Country Submissions: AFG, CMR, GMB, MDG

- * Joint Effort: BIH, COD, CZE, IRN, PSE
- * Improved Country Submissions: ARG, CHL, CUB, DEU, GHA, KHM
- * Improved Gap-filling: AGO, BEN, BFA, BWA, CAF, CIV, GIN, GNB, LBR, NAM, SLE, SSD, TGO, ZMB, ZWE, BDI, COG, GAB, GNQ, RWA, UGA
- * Improved bulk density estimation for gap-filling maps in Sub-Saharan Africa, using Random Forest prediction
- * Major inland water surfaces have been masked out from the map.
- * Corrections made in the metadata and list of contributors

VERSION 1.2.0

- * Country Submission: CHL, COL

VERSION 1.1.0 -> (Public)

- * Improved Maps: CMR, KHM
- * Country Submission: RWA

VERSION 0.14.2 -> VERSION 1.0.0 (Public)

- * Major inland water Surfaces have been masked out from the map.

VERSION 0.14

- * New Submissions: DOM, HTI
- * FRA has been added to the map (Official Submission)
- * Improved maps for ALB, BIH, VRI, CXR, FSM, HRV, MNP, NFK, PLW, SGP, SPM

VERSION 0.13b

- * FRA has been added to the map (GSP gap Filling).

VERSION 0.13a

- * Improved maps submitted by ECU, URU, PER
- * FRA has been removed from the map "official request"

VERSION 0.12.1

- * Corrected map submitted by DEU
- * Improved maps for KAZ, UZB, TKM, TJK,

VERSION 0.12

- * New country submissions: JOR, SVN
- * Improved map of BRA submitted by the country
- * Improved maps for KAZ, UZB, TKM, TJK, AFG and PAK using data provided by KAZ and UZB + WOSIS
- * Improved GSP gapfilling maps for SVK, DEU and small EU countries: AND, CYP, FRO, GGY, GIB, IMN, JEY, LIE, MCO, VAT, XAD, XNC

VERSION 0.11.1

- * New country submission: SDN
- * Joint effort: improved map IND

VERSION 0.11

- * Joint effort: IND, LAO
- * Improved map MAR

VERSION 0.10

- * Improved the procedure for filling the NA values between the country borders:
 - a 5km buffer along the boarders was used for gap-filling, excluding water bodies and coastlines;
 - no inland water bodies or urban areas were gap-filled.

VERSION 0.9.1

- * Joint effort: SYR
- * Improved maps of ETH, ARM submitted by the countries

VERSION 0.9

- * SWE Improved
- * Gapfilling using LUCAS Soil (GSP): SVK, SVN, ISL and small EU countries: AND, CYP, FRO, GGY, GIB, IMN, JEY, LIE, MCO, VAT, XAD
- * Gapfilling GSP for Carribean (BHS) and GUY, GUF
- * Joint Effort: HTI, JAM
- * New Submission: DNK, SWZ
- * Updated Map: MOZ
- * Gapfilling (SoilGrids): Small islands
- * Removed No Data zones at borderlines (GDAL, gdal_fillnodata.py, This algorithm will interpolate values for all designated nodata pixels. For each pixel a four direction conic search is done to find values to interpolate from (using inverse distance weighting).
- * Changed SoilGrids Source Data (1 km to 250 m)
- * Removed Outliers (USA, BRA, DNK)
- * Applied global mask

VERSION 0.8

- * new corrected map from PRY
- * new version of MOZ map submitted by the country
- * updated model for MLI
- * improved estimation of 0-30 stocks for ESP, IRL, FRA, GRC, BGR, ROU, LTU, LVA, POL, CZE, EST

VERSION 0.7

- * Improved bulk density estimation: CHN, BEN, BFA, CIV, GHA, GIN, GNB, LBR, MLI, NGA, SEN, SLE, TGO, BDI, CAF, COD, RWA, SSD, UGA, ZMB, ZWE

VERSION 0.5

- * FIN and TZA replaced with the country data
- * New Data: CHE, MKD, MLT
- * Calculation errors fixed: CUB, IDN, MOZ, MWI
- * Improved Model: ESP, IRL, FRA, GRC, BGR, ROU, LTU, LVA, POL, CZE, EST

* Gap Filling (SoilGrids): BGD, LAO, KHM, KOR, PRK, HND, GTM, JAM, HTI, BHS

VERSION 0.4

* Reduced size (VERSION 0.3 exported as Version 0.4 in R (raster pkg))

VERSION 0.3

* Removed reported outliers (above 2000) and minus values

VERSION 0.1

* First map combining the following 0.1 maps:

[1]	"pred/soilgrids/AFG.tif"	"pred/soilgrids/ALB.tif"
[3]	"pred/soilgrids/BIH.tif"	"pred/soilgrids/GEO.tif"
[5]	"pred/soilgrids/GUF.tif"	"pred/soilgrids/GUY.tif"
[7]	"pred/soilgrids/HRV.tif"	"pred/soilgrids/IRN.tif"
[9]	"pred/soilgrids/KGZ.tif"	"pred/soilgrids/MDG.tif"
[11]	"pred/soilgrids/MNE.tif"	"pred/soilgrids/PAK.tif"
[13]	"pred/soilgrids/PNG.tif"	"pred/soilgrids/SRBXKO.tif"
[15]	"pred/soilgrids/TJK.tif"	"pred/soilgrids/TKM.tif"
[17]	"pred/Joint-notsubmitted/ CUB.tif"	"pred/Joint-notsubmitted/ SUR.tif"
[19]	"pred/Joint-notsubmitted/ TTO.tif"	"pred/GSP/AGO.tif"
[21]	"pred/GSP/ARE.tif"	"pred/GSP/BDI.tif"
[23]	"pred/GSP/BEN.tif"	"pred/GSP/BFA.tif"
[25]	"pred/GSP/BLR.tif"	"pred/GSP/BWA.tif"
[27]	"pred/GSP/CAF.tif"	"pred/GSP/CHN.tif"
[29]	"pred/GSP/CIV.tif"	"pred/GSP/CMR.tif"
[31]	"pred/GSP/COD.tif"	"pred/GSP/COG.tif"
[33]	"pred/GSP/DZA.tif"	"pred/GSP/EGY.tif"
[35]	"pred/GSP/ERI.tif"	"pred/GSP/ESH.tif"
[37]	"pred/GSP/GAB.tif"	"pred/GSP/GHA.tif"

[39]	"pred/GSP/GIN.tif"	"pred/GSP/GNB.tif"
[41]	"pred/GSP/GNQ.tif"	"pred/GSP/IRQ.tif"
[43]	"pred/GSP/ISR.tif"	"pred/GSP/JOR.tif"
[45]	"pred/GSP/KEN.tif"	"pred/GSP/KWT.tif"
[47]	"pred/GSP/LBN.tif"	"pred/GSP/LBR.tif"
[49]	"pred/GSP/LBY.tif"	"pred/GSP/MAR.tif"
[51]	"pred/GSP/MLI.tif"	"pred/GSP/MMR.tif"
[53]	"pred/GSP/MRT.tif"	"pred/GSP/NAM.tif"
[55]	"pred/GSP/NER.tif"	"pred/GSP/NGA.tif"
[57]	"pred/GSP/OMN.tif"	"pred/GSP/QAT.tif"
[59]	"pred/GSP/RWA.tif"	"pred/GSP/SAU.tif"
[61]	"pred/GSP/SDN.tif"	"pred/GSP/SEN.tif"
[63]	"pred/GSP/SLE.tif"	"pred/GSP/SOM.tif"
[65]	"pred/GSP/SSD.tif"	"pred/GSP/SYR.tif"
[67]	"pred/GSP/TCD.tif"	"pred/GSP/TGO.tif"
[69]	"pred/GSP/TUN.tif"	"pred/GSP/TZA.tif"
[71]	"pred/GSP/UGA.tif"	"pred/GSP/YEM.tif"
[73]	"pred/GSP/ZAF.tif"	"pred/GSP/ZMB.tif"
[75]	"pred/GSP/ZWE.tif"	"pred/own/ARG.tif"
[77]	"pred/own/ARM.tif"	"pred/own/AUS.tif"
[79]	"pred/own/AUT.tif"	"pred/own/AZE.tif"
[81]	"pred/own/BEL.tif"	"pred/own/BOL.tif"
[83]	"pred/own/BRA.tif"	"pred/own/BTN.tif"
[85]	"pred/own/CAN.tif"	"pred/own/COL.tif"
[87]	"pred/own/CRI.tif"	"pred/own/DEU.tif"
[89]	"pred/own/DOM.tif"	"pred/own/ECU.tif"
[91]	"pred/own/ETH.tif"	"pred/own/GBR.tif"
[93]	"pred/own/GHA.tif"	"pred/own/HUN.tif"
[95]	"pred/own/IDN_BALI _STOCK.tif"	"pred/own/IDN_ BANTEN_STOCK.tif"
[97]	"pred/own/IDN_ GORONTALO_STOCK.tif"	"pred/own/IDN_ JABAR_STOCK.tif"
[99]	"pred/own/IDN_ JATENG_STOCK.tif"	"pred/own/IDN_ JATIM_STOCK.tif"
[101]	"pred/own/IDN_ KALBAR_STOCK.tif"	"pred/own/IDN_ KALSEL_STOCK.tif"
[103]	"pred/own/IDN_ KALTENG_STOCK.tif"	"pred/own/IDN_ KALTIM_STOCK.tif"

[105]	“pred/own/IDN_ MALUKU_STOCK.tif”	“pred/own/IDN_ MALUKU_UTARA_STOCK.tif”
[107]	“pred/own/IDN_NTB_ LOMBOK_STOCK.tif”	“pred/own/IDN_NTB_ SUMBAWA_STOCK.tif”
[109]	“pred/own/IDN_NTT_ FLORES_STOCK.tif”	“pred/own/IDN_NTT_ SUMBA_STOCK.tif”
[111]	“pred/own/IDN_NTT_ TIMOR_BARAT_STOCK.tif”	“pred/own/IDN_ PAPUA_BARAT_STOCK.tif”
[113]	“pred/own/IDN_ PAPUA_STOCK.tif”	“pred/own/IDN_ SULBAR_STOCK.tif”
[115]	“pred/own/IDN_ SULSEL_STOCK.tif”	“pred/own/IDN_ SULTENG_STOCK.tif”
[117]	“pred/own/IDN_ SULUT_STOCK.tif”	“pred/own/IDN_ SUMATERA_SOC.tif”
[119]	“pred/own/IND.tif”	“pred/own/IRQ.tif”
[121]	“pred/own/ITA.tif”	“pred/own/JOR.tif”
[123]	“pred/own/JPN.tif”	“pred/own/KAZ.tif”
[125]	“pred/own/KEN.tif”	“pred/own/LBN.tif”
[127]	“pred/own/LKA.tif”	“pred/own/LSO.tif”
[129]	“pred/own/LUX.tif”	“pred/own/MAR.tif”
[131]	“pred/own/MDA.tif”	“pred/own/MEX.tif”
[133]	“pred/own/MNG.tif”	“pred/own/MOZ.tif”
[135]	“pred/own/MWI.tif”	“pred/own/NIC.tif”
[137]	“pred/own/NLD.tif”	“pred/own/NPL.tif”
[139]	“pred/own/NZL.tif”	“pred/own/PAN.tif”
[141]	“pred/own/PER.tif”	“pred/own/PHL.tif”
[143]	“pred/own/PRY.tif”	“pred/own/RUS.tif”
[145]	“pred/own/SEN.tif”	“pred/own/SLV.tif”
[147]	“pred/own/SWE.tif”	“pred/own/THA.tif”
[149]	“pred/own/TUR.tif”	“pred/own/UKR.tif”
[151]	“pred/own/URY.tif”	“pred/own/USA_ak.tif”
[153]	“pred/own/USA_as.tif”	“pred/own/USA_conus.tif”
[155]	“pred/own/USA_hi.tif”	“pred/own/USA_pac_basin.tif”
[157]	“pred/own/USA_prvi.tif”	“pred/own/UZB.tif”
[159]	“pred/own/VEN.tif”	“pred/own/VNM.tif”

Annex C: Example scripts used in GSP gapfilling

The scripts used for the different maps prepared by the GSP Secretariat are based in the ones presented in the SOC Mapping Cookbook (Yigini *et al.*, 2018).

C1: Data preparation for soil profiles

```
dat <- read.csv(file = "data/horizons.csv")
# Explore the data
str(dat)
summary(dat)
dat_sites <- read.csv(file = "data/site-level.csv")
# Explore the data
str(dat_sites)
# summary of column CRF (Coarse
# Fragments) in the example data base
summary(dat$CRF)
# Convert NA's to 0
dat$CRF[is.na(dat$CRF)] <- 0
hist(dat$CRF)
# Creating a function in R to estimate
# BLD using the SOC SOC is the soil
# organic carbon content in n%
estimateBD <- function(SOC, method = "Saini_1996") {
```

```

OM <- SOC * 1.724
if (method == "Saini_1996") {
  BD <- 1.62 - 0.06 * OM
}
if (method == "Drew_1973") {
  BD <- 1/(0.6268 + 0.0361 * OM)
}
if (method == "Jeffrey_1979") {
  BD <- 1.482 - 0.6786 * (log(OM))
}
if (method == "Grigal_1989") {
  BD <- 0.669 + 0.941 * exp(1)^(-0.06 *
    OM)
}
if (method == "Adams_1973") {
  BD <- 100/(OM/0.244 + (100 - OM)/2.65)
}
if (method == "Honeyset_Ratkowsky_1989") {
  BD <- 1/(0.564 + 0.0556 * OM)
}
return(BD)
}

# summary of BLD (bulk density) in the
# example data base
summary(dat$BLD)
# See the summary of values produced
# using the pedo-transfer function with
# one of the proposed methods.
summary(estimateBD(dat$SOC[is.na(dat$BLD)],
  method = "Honeyset_Ratkowsky_1989"))
# Fill NA's using the pedotransfer
# function:
dat$BLD[is.na(dat$BLD)] <- estimateBD(dat$SOC[is.na(dat$BLD)],
  method = "Grigal_1989")
# explore the results
boxplot(dat$BLD)
# Load aqp package
library(aqp)

```

```

# Promote to SoilProfileCollection The
# SoilProfileCollection is a object
# class in R designed to handle soil
# profiles
depths(dat) <- ProfID ~ top + bottom
# Merge the soil horizons information
# with the site-level information from
# dat_sites
site(dat) <- dat_sites
# Set spatial coordinates
coordinates(dat) <- ~X + Y
# A summary of our
# SoilProfileCollection
dat

library(GSIF)
## Estimate 0-30 standard horizon using
## mass preserving splines
try(SOC <- mpspline(dat, "SOC", d = t(c(0,
30))))
try(BLD <- mpspline(dat, "BLD", d = t(c(0,
30))))
try(CRFVOL <- mpspline(dat, "CRF", d = t(c(0,
30))))
## Prepare final data frame
dat <- data.frame(id = dat@site$ProfID, Y = dat@sp@coords[,
2], X = dat@sp@coords[, 1], SOC = SOC$var.std[,
1], BLD = BLD$var.std[, 1], CRFVOL = CRFVOL$var.std[,
1])

dat <- dat[complete.cases(dat), ]
## Take a look to the results
head(dat)
# Estimate Organic Carbon Stock SOC
# must be in g/kg BLD in kg/m3 CRF in
# percentage
OCSKGM <- OCSKGM(ORCDRC = dat$SOC, BLD = dat$BLD *
1000, CRFVOL = dat$CRFVOL, HSIZE = 30)

```

```

dat$OCSKGM <- OCSKGM
dat$meaERROR <- attr(OCSKGM, "measurementError")
dat <- dat[dat$OCSKGM > 0, ]
summary(dat)
## We can save our processed data as a
## table
write.csv(dat, "data/dataproc.csv")

```

C2: Mixing covariates and soil points data

```

# Load the processed data. This table
# was prepared in the previous chapter.
dat <- read.csv("data/dataproc.csv")
files <- list.files(path = "covs", pattern = "tif$",
  full.names = TRUE)
covs <- stack(files)
covs <- stack(covs, soilmap.r)
# correct the name for layer 14
names(covs)[14] <- "soilmap"
# mask the covariates with the country
# mask from the data repository
mask <- raster("data/mask.tif")
covs <- mask(x = covs, mask = mask)

plot(covs)
# upgrade points data frame to
# SpatialPointsDataFrame
coordinates(dat) <- ~X + Y
# extract values from covariates to the
# soil points
dat <- extract(x = covs, y = dat, sp = TRUE)
# LCEE10 and soilmap are categorical
# variables
dat@data$LCEE10 <- as.factor(dat@data$LCEE10)
dat@data$soilmap <- as.factor(dat@data$soilmap)

```

```
# levels(soilmap) <- Symbol.levels
summary(dat@data)
dat <- as.data.frame(dat)
# The points with NA values has to be
# removed
dat <- dat[complete.cases(dat), ]
# export as a csv table
write.csv(dat, "data/MKD_RegMatrix.csv",
          row.names = FALSE)
```

C3: Fitting a RK model to predict the OCS

```
# load data
dat <- read.csv("data/MKD_RegMatrix.csv")
dat$LCEE10 <- as.factor(dat$LCEE10)
dat$soilmap <- as.factor(dat$soilmap)
# explore the data structure
str(dat)
library(sp)
# Promote to spatialPointsDataFrame
coordinates(dat) <- ~ X + Y
class(dat)
dat@proj4string <- CRS(projargs = "+init=epsg:4326")
dat@proj4string
library(raster)
# list all the itf files in the folder covs/
files <- list.files(path = "covs", pattern = "tif$",
                    full.names = TRUE)
# load all the tif files in one rasterStack object
covs <- stack(files)
# load the vectorial version of the soil map
soilmap <- shapefile("MK_soilmap_simple.shp")
# rasterize using the Symbol layer
soilmap@data$Symbol <- as.factor(soilmap@data$Symbol)
soilmap.r <- rasterize(x = soilmap, y = covs[[1]],
```

```
field = "Symbol")
# stack the soil map and the other covariates
covs <- stack(covs, soilmap.r)
# correct the name for layer 14
names(covs)[14] <- "soilmap"
# print the names of the 14 layers:
names(covs)
datdf <- dat@data
datdf <- datdf[, c("OCSKGM", names(covs))]
## Fit a multiple linear regression model
## between the log transformed values
## of OCS and the top 20 covariates
model.MLR <- lm(log(OCSKGM) ~ ., data = datdf)
## stepwise variable selection
model.MLR.step <- step(model.MLR, direction="both")
## summary and anova of the new model
## using stepwise covariates selection
summary(model.MLR.step)
anova(model.MLR.step)
## graphical diagnosis of the regression analysis
par(mfrow=c(2,2))
plot(model.MLR.step)
par(mfrow=c(1,1))
## collinearity test using variance inflation factors
library(car)
vif(model.MLR.step)
#problematic covariates should have sqrt(VIF) > 2

sqrt(vif(model.MLR.step))
## Removing B07CHE3 from the stepwise model:
model.MLR.step <- update(model.MLR.step, . ~ . - B07CHE3)
# Test the vif again:
sqrt(vif(model.MLR.step))
## summary of the new model using stepwise covariates selection
summary(model.MLR.step)
# outlier test using the Bonferroni test
outlierTest(model.MLR.step)
# Project point data.
```

```

dat <- spTransform(dat, CRS("+init=epsg:6204"))
# project covariates to VN-2000 UTM 48N
covs <- projectRaster(covs,
                      crs = CRS("+init=epsg:6204"), method='ngb')
covs$LCEE10 <- as.factor(covs$LCEE10)
covs$soilmap <- as.factor(covs$soilmap)

## Promote covariates to spatial grid dataframe.
covs.sp <- as(covs, "SpatialGridDataFrame")
covs.sp$LCEE10 <- as.factor(covs.sp$LCEE10)
covs.sp$soilmap <- as.factor(covs.sp$soilmap)
### RK model
library(automap)
## Run regression kriging prediction.
## This step can take hours...!
OCS.krige <- autoKrige(
  formula = as.formula(model.MLR.step$call$formula),
  input_data = dat,
  new_data = covs.sp,
  verbose = TRUE,
  block = c(1000, 1000))
OCS.krige
## Convert prediction and standard deviation to rasters
## And back-transform the vlaues
RKprediction <- exp(raster(OCS.krige$krige_output[1]))
RKpredsd <- exp(raster(OCS.krige$krige_output[3]))
plot(RKprediction)

## Save results as tif files
writeRaster(RKprediction,
            filename = "results/MKD_OCSKGM_RK.tif")
writeRaster(RKpredsd,
            filename = "results/MKD_OCSKGM_RKpredsd.tif")
# save the model
saveRDS(model.MLR.step, file="results/RKmodel.Rds")

```

C4: Fitting a random forest model to predict the OCS

```
library(reshape)
# Correlation analysis to select
# covariates
names(dat)
COR <- cor(as.matrix(dat[, 7]), as.matrix(dat[,
  -c(1:8)]))
COR
x <- subset(melt(COR), value != 1 | value !=
  NA)
x <- x[with(x, order(-abs(x$value))), ]
x[1:25, ]
idx <- as.character(x$X2[1:25])
dat2 <- dat[c("OCSKGM", idx)]
names(dat2)
COVall <- COV
COV <- COV[[idx]]
plot(COV)
library(randomForest)
# Try different values of mtry and
# select the model with the optimal
# value
model <- tuneRF(dat[, c(names(COV))], dat$OCSKGM,
  stepFactor = 1.5, doBest = TRUE, improve = 0.5)
# Use the model to predict the SOC in
# the covariates space
beginCluster()
start <- Sys.time()
pred <- clusterR(COV, predict, args = list(model))
print(Sys.time() - start)
endCluster()
```

C5: Fitting a svm model to predict the OCS

```

# Correlation analysis to select
# covariates
names(dat)
COR <- cor(as.matrix(dat[, 7]), as.matrix(dat[,
  -c(1:8)]))
COR
x <- subset(melt(COR), value != 1 | value !=
  NA)
x <- x[with(x, order(-abs(x$value))), ]
x[1:25, ]
idx <- as.character(x$X2[1:25])
dat2 <- dat[c("OCSKGM", idx)]
names(dat2)
COVall <- COV
COV <- COV[[idx]]
plot(COV)
library(e1071)
library(caret)
# Test different values of epsilon and
# cost
tuneResult <- tune(svm, OCSKGM ~ ., data = dat[,
  c("OCSKGM", names(COV))], ranges = list(epsilon = seq(0,
  1, 0.1), cost = c(0.5, 1, 1.5, 2, 5,
  10)))
# Choose the model with the best
# combination of epsilon and cost
tunedModel <- tuneResult$best.model
# Use the model to predict the SOC in
# the covariates space
beginCluster()
start <- Sys.time()
pred <- clusterR(COV, predict, args = list(tunedModel))
print(Sys.time() - start)
endCluster()

```

Annex D: GSOCmap submission overview

This sections lists the data specifications, methodological specifications, and contacts for each country.

D1: Afghanistan

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 384

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Measured

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest in R

Validation: $R^2=48\%$

Contact Point:

Data holder: Hameedullah Ahmadzai

Institution: FAO Afghanistan

Email or website: Hameedullah.Ahmadzai@fao.org

D2: Albania**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D3: Algeria**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 16

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2029

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D4: Andorra**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: NA

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: N/A

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D5: Angola**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 962

Time frame: before and after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2017

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D6: Antigua and Barbuda**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D7: Argentina**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 5073

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: under review

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest + QRF

Mapping method details: Random forest + QRF

Validation: ME: -0.48 t/ha; RMSE: 17.23 t/ha, AVE: 0.37

Contact Point:

Data holder: Guillermo Federico Olmedo

Institution: Instituto Nacional de Tecnología Agropecuaria (INTA)

Email or website: olmedo.guillermo@inta.gob.ar

D8: Armenia

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 40

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Other

Mapping method details: linear spectral unmixing pixel based classification

Validation: N/A

Contact Point:

Data holder: Sahakyan Samvel

Institution: Soil Science, Melioration and Agrochemistry Scientific Center
named after H. Petrosyan

Email or website: ssahakyan@yandex.ru

D9: Australia

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 5588

Time frame: after 1990

Soil Organic Carbon method: mixed

Bulk density method: mixed

Coarse fragments method: Measured

Citation: Viscarra Rossel RA, Webster R, Bui EN, Baldock JA (2014). Baseline map of organic carbon in Australian soil to support national carbon accounting and monitoring under climate change. *Global Change Biology* 20, 2953-2970. doi: 10.1111/gcb.12569

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: Combination of decision trees with piecewise regression on environmental variables and geostatistical modelling of residuals.

Validation: Refer to Viscarra Rossel et al. (2014) for detailed analysis of errors and confidence intervals. The total stock of organic C in the 0–30 cm layer of soil for Australia is 24.97 Gt with 95 percent confidence limits of 19.04 and 31.83 Gt. See maps of the 5 percent and 95 percent confidence limits for geographical variation across the continent. Further information on errors can be provided on request.

Contact Point:

Data holder: Mike Grundy

Institution: CSIRO Agriculture and Food

Email or website: mike.grundy@csiro.au

D10: Austria**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 150511

Time frame: before and after 1990

Soil Organic Carbon method: mixed

Bulk density method: Pedotransfer functions

Coarse fragments method: Estimated

Citation: FBVA (Ed.) "Österreichische Waldboden-Zustandsinventur". Mitteilungen der Forstlichen Bundesversuchsanstalt 168 (1992).

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Mixed

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Federal research and training center for forest, natural hazards and landscape Austria

Institution: Dpt. for forest ecology and soil

Email or website: michael.englisch@bfw.gv.at

D11: Azerbaijan**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 430

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Measured

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Amin Ismayilov

Institution: Institute of Soil Science and Agrochemistry of ANAS

Email or website: amin_ismayilov@mail.ru; amin.ismayil@gmail.com

D12: Bahamas

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: NA

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: N/A

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Other

Mapping method details: Data extracted from the model for Central America based on WOSIS data

Validation: N/A

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D13: Bahrain**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D14: Bangladesh**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D15: Barbados**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D16: Belarus**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 88

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2021

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines

Validation: MaE=45.3 t/ha; RMSE=64 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D17: Belgium

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 2457

Time frame: after 1990

Soil Organic Carbon method: mixed

Bulk density method: mixed

Coarse fragments method: Measured

Citation: C. Chartin, S. Lettens, P. Verschelde, S. Buyle, K. Oorts, P. Engels,

M. Swerts, B. De Vos, B. van Wesemael, 2017. The Belgian contribution to the Global Soil Organic Carbon Stock map, Proceedings of Soil Resources Mapping: past, present and future, Thematic Day 2017 of the Soil Science Society of Belgium. Spatial analysis of soil organic carbon evolution in Belgian croplands and grasslands, 1960-2006. *Global Change Biology* 17(1): 466-479.

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: Agricultural land (Flanders): The following empirical regression model was derived based on a dataset of 352 profiles. $\%SOC = \text{LandUse} + a.\text{clay} + b.H_2O_{min} + c.LandUse.Clay + d.LandUse.H_2O_{min}$ Based on the Belgian soil map and the VITO land use map (Poelmans, 2014), this regression equation was applied to the entire territory of Flanders. Agricultural land (Wallonia): i/A Generalized Additive Model (GAM; Wood, 2001) was fitted on 2/3 of the dataset. Spatialized environmental covariates (40m x 40m) were used as inputs on the model to map SOC stocks over croplands and grasslands in Wallonia (Southern Belgium).; Forest (Flanders): The average soil carbon stock in the upper 30 cm (C_s , in t C/ha) is computed per texture-drainage class of the Belgian soil map. This value is pasted into the 10x10 m² grid of the land use map. ; Forest (Wallonia): i/A Generalized Additive Model (GAM; Wood, 2001) was fitted on 2/3 of the dataset. Spatialized environmental covariates (40m x 40m) were used as inputs on the model to map SOC stocks over forest in Wallonia (Southern Belgium).

Validation: Agricultural land (Flanders): The uncertainty reported is the model uncertainty on point estimates for each data point, in which the estimated model parameters are simulated 1000 times, under the assumption that they are independent and normally distributed variables, using their model estimation and standard error as distribution parameters. (Goidts, 2009 and Meersmans, 2011); Agricultural land (Wallonia): The external validation (on the remaining 1/3 of the dataset) gave a R^2 of 0.64 and a RMSE of 16 Mg C / ha. ii/ The computation of the prediction uncertainty accounts for the errors associated to both the estimations of i) SOC stocks and ii) parameters of the spatial model (GAM). Hence, two consecutive stochastic simulations (Monte-Carlo method) were used to produce 10,000 (i.e., 100 x 100) independent spatialized datasets. Based on these 10,000 individuals, mean SOC stocks and standard deviation (SD) were computed for each pixel. (Chartin et al., 2017) Forest (Flanders): The uncer-

tainty of the mean (precision) is based on the margin of error (ME) derived from half the 95% confidence interval (CI95%). CI95% are estimated based on bias corrected and accelerated (BCa) percentiles at 2.5 and 97.5% determined by bootstrapping (B = 5000 resamples). Forest (Wallonia): The external validation (on the remaining 1/3 of the dataset) gave a R² of 0.41, a mean error of 0.3 Mg C /ha, a MAE of 16 Mg C /ha and a RMSE of 18.2 Mg C / ha. ii/ The computation of the prediction uncertainty (standard deviation, SD) accounts only for the errors associated to the estimation of the parameters of the spatial model (GAM). The mgcv package in R provides a Bayesian approach to compute standard errors for the predictions (Wood, 2001).

Contact Point:

Data holder: Vlaamse overheid and Service Public de Wallonie
 Institution: Data Holders: Vlaamse overheid and Service Public de Wallonie;
 Data Handlers: 1) Georges Lemaitre Centre for Earth and Climate Research, Earth and Life Institute, Universite Catholique de Louvain, 1348 Louvain-la-Neuve, Belgium 2) Environment and Climate unit, Research Institute for Nature and Forest, 1070 Brussels, Belgium 3) Vlaams Planbureau voor Omgeving, Departement Omgeving, Vlaamse overheid, 1000 Brussel, Belgium 4) Service Public de Wallonie, Direction Generale de l'Agriculture, des Ressources Naturelles et de l'Environnement (DGO3), 5100 Namur, Belgium
 Email or website: Flanders: katrien.oorts@vlaanderen.be; Wallonia: patrick.engels@spw.wallonie.be

D18: Belize

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 20

Time frame: external datasets

Soil Organic Carbon method: external datasets

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: external datasets

Citation: Guevara, M., Olmedo, G. F., Stell, E., Yigini, Y., Aguilar Duarte, Y., Arellano Hernández, C., Arévalo, G. E., Arroyo-Cruz, C. E., Bolivar, A., Bunning, S., Bustamante Cañas, N., Cruz-Gaistardo, C. O., Davila, F., Dell Acqua, M., Encina, A., Figueredo Tacona, H., Fontes, F., Hernández Herrera, J. A., Ibelles Navarro, A. R., Loayza, V., Manueles, A. M., Mendoza Jara, F., Olivera, C., Osorio Hermosilla, R., Pereira, G., Prieto, P., Ramos, I. A., Rey Brina, J. C., Rivera, R., Rodríguez-Rodríguez, J., Roopnarine, R., Rosales Ibarra, A., Rosales Riveiro, K. A., Schulz, G. A., Spence, A., Vasques, G. M., Vargas, R. R., and Vargas, R.: No silver bullet for digital soil mapping: country-specific soil organic carbon estimates across Latin America, *SOIL*, 4, 173–193, <https://doi.org/10.5194/soil-4-173-2018>, 2018.

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: N/A

Validation: $R^2=0.33$; $MAE=15.88$; $RMSE=31.28$.

Contact Point:

Data holder: Global Soil Partnership

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

D19: Benin

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 714

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2020

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D20: Bhutan**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 993
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Measured
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Mixed
Mapping method details: Regression tree using cubist and R Kriging using Vesper
Validation: ME 0.5; RMSE 16.3; R2 0.63

Contact Point:

Data holder: Tsheten Dorji and Dr Tshering Dorji
Institution: National Soil Services Centre, Department of Agriculture, Ministry of Agriculture and Forests
Email or website: tshetendorji08@gmail.com and tsericdoji@gmail.com

D21: Bolivia (Plurinational State of)**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 4788
Time frame: before and after 1990

Soil Organic Carbon method: mixed
Bulk density method: Pedotransfer functions
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Random forest
Mapping method details: Random forest
Validation: $R^2 = 0.287$

Contact Point:

Data holder: Hernan Figueredo Ticona
Institution: Viceministerio de Tierras
Email or website: hernan.figueredo@yahoo.com

D22: Bosnia and Herzegovina

GSOCmap layer source:

Map source: Joint effort with GSP

Data specifications:

Number of samples: 1715
Time frame: before and after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Measured
Coarse fragments method: Measured
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: MAE=15.2 t/ha; RMSE=17.8 t/ha;

Contact Point:

Data holder: Prof. dr. Hamid Custovic

Institution: University of Sarajevo

Email or website: custovic.hamid@gmail.com

D23: Botswana**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 839

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2022

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D24: Brazil

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 6998

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: Ensemble model combining nine methods (stepwise multiple linear regression, elastic net, principal components regression, partial

least squares regression, multivariate adaptive regression splines, cubist, regression tree, random forest and extreme gradient boosting)

Validation: Training: ME = 1.55 t/ha, RMSE = 21.93 t/ha; Validation: ME = 5.82 t/ha, RMSE = 54.05 t/ha

Contact Point:

Data holder: Gustavo M. Vasques

Institution: Embrapa Solos

Email or website: gustavo.vasques@embrapa.br

D25: Brunei Darussalam

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D26: Bulgaria

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 664

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D27: Burkina Faso**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 532

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2021

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D28: Burundi**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 34

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2019

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Burundi, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Rwanda, Uganda

Validation: MAE=16 t/ha; RMSE=19.1 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D29: Cambodia**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 671
Time frame: after 1990
Soil Organic Carbon method: dry combustion
Bulk density method: Pedotransfer functions
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Regression Kriging
Mapping method details: Regression Kriging
Validation: N/A

Contact Point:

Data holder: Dr. Koy Ra
Institution: DALRM, General Directorate of Agriculture
Email or website: koyra2010@yahoo.com

D30: Cameroon**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 1250
Time frame: before and after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: N/A

Contact Point:

Data holder: Francis B. T. Silatsa and Martin Yemefack

Institution: University of Dschang, Department of Soil Science; International Institute of Tropical Agriculture (IITA), Yaounde

Email or website: silatsat@yahoo.fr; myemefack@yahoo.fr

D31: Canada

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 48017

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: ensemble map from 11 contributions using conventional upscaling, RF, and other algorithms

Validation: We provide a sd based on the variation between overlapping contributions. Error rates for the individual contributions varied from none (polygon averages), low reliability for some maps (10% concordance), and some had better results (30 % or higher). Final map reliability estimates await the development of a national validation dataset (in progress).

Contact Point:

Data holder: Bert VandenBygaart

Institution: Agriculture and Agri-Food Canada

Email or website: bert.vandenbygaart@agr.gc.ca

D32: Cape Verde**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D33: Central African Republic

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 83

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2023

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D34: Chad

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 5

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2063

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D35: Chile

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 3258

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: $R^2=0.34$

Contact Point:

Data holder: Rodrigo Osorio Hermosilla

Institution: Servicio Agrícola y Ganadero, Departamento De Suelos

Email or website: rodrigo.osorio@sag.gob.cl

D36: China**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 1487

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2025

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines

Validation: MAE=32.3 t/ha; RMSE=49.2 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D37: Colombia**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 4329

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: NA

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Regression Kriging

Mapping method details: Regression-kriging spatial interpolation technique that combines a regression of the dependent variable (target variable) over the predictors (i.e., the environmental covariates) with kriging of the prediction residuals.

Validation: ME: 0.0006705, MAE: 0.5582, RMSE: 0.7416, R2 : 0.5843

Contact Point:

Data holder: German Dario Alvarez Lucero

Institution: Instituto Geografico Agustin Codazzi

Email or website: german.alvarez@igac.gov.co

D38: Comoros

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D39: Congo

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 68

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2027

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Burundi, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Rwanda, Uganda

Validation: MAE=16 t/ha; RMSE=19.1 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D40: Cook Islands

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D41: Costa Rica

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 1518

Time frame: before and after 1990

Soil Organic Carbon method: mixed

Bulk density method: mixed

Coarse fragments method: No data

Citation: NA

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: MAE = 28.8 t/ha, RMSE=34.9 t/ha, R²=0.265

Contact Point:

Data holder: Alban Rosales Ibarra/Bryan Aleman Montes
Institution: Instituto Nacional de Innovacion de Transferencia en Tecnologia (INTA); Centro de investigaciones Agronomicas, Universidad de Costa Rica
Email or website: arosaarosales@inta.go.cr/bryan.aleman@ucr.ac.cr

D42: Côte d'Ivoire**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 250
Time frame: before 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: mixed
Coarse fragments method: No data
Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2026

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Random forest
Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe
Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R2=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D43: Croatia**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D44: Cuba**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 30886

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: ME=-5 t/ha; MAE=16.44 t/ha; RMSE=23.74 t/ha; AVE=0.37

Contact Point:

Data holder: Dr. Luis A. Gomez Jorin (General Director), Dr. Luis B. Rivero Ramos (Project Leader), Roberto Morales

Institution: Instituto de Suelos, Ministerio de Agricultura

Email or website: director@isuelos.co.cu; roberto.morales@isuelos.co.cu

D45: Cyprus**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 90

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LU-CAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D46: Czech Republic**GSOCmap layer source:**

Map source: Joint effort with GSP

Data specifications:

Number of samples: 4169

Time frame: before and after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Measured

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: MAE=13.3 t/ha; RMSE=14.8 t/ha; R2=0.29

Contact Point:

Data holder: Josef Kozak

Institution: Czech University of Life Sciences Prague

Email or website: kozak@af.czu.cz

D47: Democratic People's Republic of Korea**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D48: Democratic Republic of the Congo**GSOCmap layer source:**

Map source: Joint effort with GSP

Data specifications:

Number of samples: 773

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Burundi, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Rwanda, Uganda

Validation: MAE=16 t/ha; RMSE=19.1 t/ha

Contact Point:

Data holder: Eric Lutete Landu ; Yves-Dady Botula Manyala

Institution: Observatoire Satellital des Forêts d'Afrique Centrale (OSFAC) /Université de Kinshasa (Unikin)

Email or website: ericlutete@gmail.com ; ydbotula@yahoo.fr

D49: Denmark**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 46850

Time frame: before and after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Measured

Coarse fragments method: No data

Citation: Adhikari K, Hartemink AE, Minasny B, Bou Kheir R, Greve MB, et al. (2014) Digital Mapping of Soil Organic Carbon Contents and Stocks in Denmark. PLoS ONE 9(8): e105519. [doi:10.1371/journal.pone.0105519](https://doi.org/10.1371/journal.pone.0105519)

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Regression Kriging

Mapping method details: Regression kriging

Validation: For the 0-5 cm layer, the mean error was 1.1 g/kg

Contact Point:

Data holder: Mogens H Greve, Mette B Greve

Institution: Agroecology, Aarhus University

Email or website: mogensh.greve@agro.au.dk

D50: Djibouti**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2028

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen
Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D51: Dominica**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D52: Dominican Republic**GSOCmap layer source:**

Map source: Joint effort with GSP

Data specifications:

Number of samples: 120
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Pedotransfer functions
Coarse fragments method: Estimated
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Random forest
Mapping method details: N/A
Validation: N/A

Contact Point:

Data holder: Rafael Antonio Rivera
Institution: Ministerio De Medio Ambiente y Recursos Naturales
Email or website: rafantoniorive@gmail.com

D53: Ecuador**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 12861
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: mixed

Coarse fragments method: Estimated
Citation: NA

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Regression Kriging
Mapping method details: R Kriging
Validation: ME: 0.0016; MAE:0.396; RMSE: 0.534; R2: 0.628

Contact Point:

Data holder: Veronica Loayza, Wilmer Jimenez
Institution: Ministerio de Agricultura y Ganaderia del Ecuador
Email or website: veronica_loayza@yahoo.es/wjimenez@mag.gob.ec/nloayza@mag.gob.ec

D54: Egypt

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 22
Time frame: before and after 1990
Soil Organic Carbon method: mixed
Bulk density method: mixed
Coarse fragments method: Measured/Estimated
Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2030

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D55: El Salvador**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 866

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Sol Munoz/ Rene Arevalo

Institution: Ministerio de Medio Ambiente y Recursos Naturales (MARN)/
Centro Nacional de Tecnologia Agropecuaria y Forestal (CENTA)

Email or website: smunoz@marn.gob.sv / rene.arevalo@centa.gob.sv

D56: Equatorial Guinea

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2037

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally

available data from Burundi, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Rwanda, Uganda

Validation: MAE=16 t/ha; RMSE=19.1 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D57: Eritrea

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2031

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon,

Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen
Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D58: Estonia**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 220

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LU-CAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D59: Eswatini

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 91

Time frame: after 1990

Soil Organic Carbon method: mixed

Bulk density method: mixed

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest regression

Validation: RMSE 14.25 t/ha, R2 0.76

Contact Point:

Data holder: Dr. Wisdom M. Dlamini

Institution: Department of Geography, Environmental Science and Planning,
University of Eswatini

Email or website: mwdlamini@gmail.com, wdlamini@uniswa.sz

D60: Ethiopia**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 58957

Time frame: after 1990

Soil Organic Carbon method: soil spectroscopy

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: ME = 0.04104367, RMSE = 18.7978, and R² = 0.5431023

Contact Point:

Data holder: Kiflu Gudeta

Institution: Ministry of Agriculture and Natural Resources

Email or website: gkiflu@gmail.com

D61: Faroe Islands

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: N/A

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D62: Fiji

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D63: Finland

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 2237

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: mixed

Coarse fragments method: Measured

Citation: Biosoil: <https://doi.org/10.5194/gmd-9-4169-2016>, 2016, Lucas: <https://doi.org/10.1016/j.geoderma.2015.07.006>, Soil Database: [https://doi.org/10.1016/S0166-2481\(06\)31005-7](https://doi.org/10.1016/S0166-2481(06)31005-7)

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: Random forest

Validation: N/A

Contact Point:

Data holder: Harri Lilja

Institution: Natural Resources Institute Finland

Email or website: harri.lilja@luke.fi

D64: France

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 2952

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: N/A

Citation: France carbon soil map - INRA (with data provided by GIS Sol) - M. Martin, D. Arrouays, V.L. Mulder, M. Lacoste, A.C.Richer-de-Forges, M. Bardy and A. Bispo. 2017 - Updated 1/12/2017 Mulder, V.L., Lacoste, M., Richer-de-Forges, A.C., Martin, M.P., Arrouays, D., 2016. National versus global modelling the 3D distribution of soil organic carbon in mainland France. Geoderma 263, 16 - 34. <https://doi.org/10.1016/j.geoderma.2015.08.035>

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Other

Mapping method details: Cubist

Validation: N/A

Contact Point:

Data holder: Manuel Martin

Institution: Institut National de la Recherche Agronomique (INRA)

Email or website: manuel.martin@inra.fr

D65: Gabon**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 46

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2033

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Burundi, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Rwanda, Uganda

Validation: MAE=16 t/ha; RMSE=19.1 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D66: Gambia**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 24

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Measured

Coarse fragments method: Measured

Citation: NA

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: N/A

Contact Point:

Data holder: Abdou Rahman Jobe

Institution: Soil and Water Management Services Unit, Ministry of Agriculture

Email or website: armjobe@yahoo.com; samatehala@yahoo.co.uk

D67: Georgia

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 0

Time frame: external datasets

Soil Organic Carbon method: external datasets

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: external datasets

Citation: N/A

Methodological specifications:

Mapping technique: external datasets

Mapping method: external datasets

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Global Soil Partnership

Institution: N/A

Email or website: N/A

D68: Germany

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 4863

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Measured

Coarse fragments method: Measured

Citation: <https://onlinelibrary.wiley.com/doi/full/10.1111/gcb.12558> https://www.thuenen.de/media/institute/ak/Allgemein/news/Thuenen_Report_64_final.pdf https://literatur.thuenen.de/digbib_extern/dn059837.pdf https://literatur.thuenen.de/digbib_extern/dn057211.pdf

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Erik Gruneberg, Nicole Wellbrock

Institution: Thunen-Institute of Forests Ecosystems, Thunen-Institutes of Climate-Smart Agriculture

Email or website: erik.gruneberg@thuenen.de, nicole.wellbrock@thuenen.de

D69: Ghana**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 751
Time frame: before and after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Pedotransfer functions
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Random forest
Mapping method details: Random forest
Validation: ME=-2.5 t/ha; MAE=12 t/ha, RMSE =15 t/ha; AVE = 0.18

Contact Point:

Data holder: Stephen Owusu
Institution: CSIR-Soil Research Institute, Kwadaso-Kumasi, Ghana.
Email or website: s.owusu@csir-soilresearch.org ; stephenowusu41@yahoo.com

D70: Greece**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 491
Time frame: after 1990
Soil Organic Carbon method: dry combustion
Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D71: Grenada

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D72: Guatemala

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 15

Time frame: external datasets

Soil Organic Carbon method: external datasets

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: external datasets

Citation: Guevara, M., Olmedo, G. F., Stell, E., Yigini, Y., Aguilar Duarte, Y., Arellano Hernández, C., Arévalo, G. E., Arroyo-Cruz, C. E., Bolivar, A., Bunning, S., Bustamante Cañas, N., Cruz-Gaistardo, C. O., Davila, F., Dell Acqua, M., Encina, A., Figueredo Tacona, H., Fontes, F., Hernández Herrera, J. A., Ibelles Navarro, A. R., Loayza, V., Manueles, A. M., Mendoza Jara, F., Olivera, C., Osorio Hermosilla, R., Pereira, G., Prieto, P., Ramos, I. A., Rey Brina, J. C., Rivera, R., Rodríguez-Rodríguez, J., Roopnarine, R., Rosales Ibarra, A., Rosales Riveiro, K. A., Schulz, G. A., Spence, A., Vasques, G. M., Vargas, R. R., and Vargas, R.: No silver bullet for digital soil mapping: country-specific soil organic carbon estimates across Latin America, *SOIL*, 4, 173–193, <https://doi.org/10.5194/soil-4-173-2018>, 2018.

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: N/A

Validation: R²=0.33; MAE=15.88; RMSE=31.28.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D73: Guinea

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 61

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2035

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D74: Guinea-Bissau

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 17

Time frame: before 1990

Soil Organic Carbon method: N/A

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2036

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D75: Guyana

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 43

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2039

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: ensemble of different SVM models based on 238 points from the region

Validation: N/A

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D76: Haiti

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 135

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2040

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Other

Mapping method details: Bayesian Regression where the response variable is observational data from Haiti and the explanatory

Validation: N/A

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D77: Honduras

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 7

Time frame: external datasets

Soil Organic Carbon method: external datasets

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: external datasets

Citation: Guevara, M., Olmedo, G. F., Stell, E., Yigini, Y., Aguilar Duarte, Y., Arellano Hernández, C., Arévalo, G. E., Arroyo-Cruz, C. E., Bolivar, A., Bunning, S., Bustamante Cañas, N., Cruz-Gaistardo, C. O., Davila, F., Dell Acqua, M., Encina, A., Figueredo Tacona, H., Fontes, F., Hernández Herrera, J. A., Ibelles Navarro, A. R., Loayza, V., Manueles, A. M., Mendoza Jara, F., Olivera, C., Osorio Hermosilla, R., Pereira, G., Prieto, P., Ramos, I. A., Rey Brina, J. C., Rivera, R., Rodríguez-Rodríguez, J., Roopnarine, R., Rosales Ibarra, A., Rosales Riveiro, K. A., Schulz, G. A., Spence, A., Vasques, G. M., Vargas, R. R., and Vargas, R.: No silver bullet for digital soil mapping: country-specific soil organic carbon estimates across Latin America, *SOIL*, 4, 173–193, <https://doi.org/10.5194/soil-4-173-2018>, 2018.

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: N/A

Validation: $R^2=0.33$; $MAE=15.88$; $RMSE=31.28$.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D78: Hungary

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 1236

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Measured

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Quantile regression forest

Validation: ME= -1.72 t/ha, MAE= 17.08 t/ha, RMSE= 23.18 t/ha

Contact Point:

Data holder: Lazlo Pasztor

Institution: Institute for Soil Sciences and Agricultural Chemistry, Centre for Agricultural Research, Hungarian Academy of Sciences

Email or website: pasztor@rissac.hu

D79: Iceland

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: N/A

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D80: India**GSOCmap layer source:**

Map source: Joint effort with GSP

Data specifications:

Number of samples: 175993
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: mixed
Coarse fragments method: Estimated
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Support vector machines
Mapping method details: Support vector machine
Validation: ME 0.002269 t/ha, RMSE 8.383869 t/ha, R2 0.53

Contact Point:

Data holder: S. K. Singh, A.K. Patra
Institution: National Bureau of Soil Survey and Land Use Planning, Nagpur;
Indian Institute of soil Science, Bhopal
Email or website: skcssri@gmail.com, patraak@gmail.com

D81: Indonesia**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 15750
Time frame: before and after 1990
Soil Organic Carbon method: wet oxidation

Bulk density method: Measured

Coarse fragments method: No data

Citation: Sulaeman et al. 2012 Sulaeman et al. 2013, 2014, 2015, 2016

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: YIYI SULAEMAN

Institution: Indonesian Center for Agricultural Land Resource Research and Development

Email or website: y.sulaeman@gmail.com

D82: Iran (Islamic Republic of)

GSOCmap layer source:

Map source: Joint effort with GSP

Data specifications:

Number of samples: 954

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: $R^2 = 0.58$, RMSE=18.5 t/ha

Contact Point:

Data holder: Kambiz Bazargan

Institution: Soil and Water Research Institute (SWRI)

Email or website: bazargan_k@yahoo.com

D83: Iraq**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 400

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Dr Eman

Institution: Ministry of Agriculture

Email or website: eman_sahib@yahoo.com

D84: Ireland**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 233

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership
Institution: European Soil Data Centre (ESDAC)
Email or website: esdac@jrc.ec.europa.eu

D85: Israel**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 12
Time frame: before 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Pedotransfer functions
Coarse fragments method: Measured/Estimated
Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2041

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Support vector machines
Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen
Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D86: Italy**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 6748

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: mixed

Coarse fragments method: Measured

Citation: SCS map of Italy partnership: structure and contributions Coordinator: Edoardo Costantini CREA Firenze. WG1 Data, metadata and covariates supplying: CREA-Firenze; CREA-Roma; ISPRA (covariates); IPLA, Piemonte; ERSAP, Lombardia; ARPA Veneto; ERSAP-FVG, Friuli Venezia-Giulia; Regione Liguria; SGSS, Emilia-Romagna; Consorzio Lamma, Toscana; Regione Marche; Regione Campania; Regione Puglia, Università di Foggia, Università Mediterranea di Reggio Calabria, ARSSA, Calabria; Regione Sicilia. WP2 Harmonization of SOC analytical methods and bulk density values and estimation a) SOC methods: ARPA Veneto Università Mediterranea di Reggio Calabria; Università di Foggia; CREA-Roma; CNR-ISE, Firenze. b) Bulk density estimation: CNR-Ibimet; Firenze; CREA, Firenze; ARPA Veneto; Consorzio Lamma, Firenze; SGSS, Emilia-Romagna. WP3 Digital soil mapping: Methods: CNR-Ibimet, Firenze; CREA Firenze; CREA Roma. Quality checking: ARPA Veneto; SGSS, Emilia-Romagna. WP4-elaboration and management of data and maps: CREA Firenze; CNR-Ibimet, Firenze. Authors (alphabetic order for institutions and authors) Paolo Giandon, Ialina Vinci (ARPA Veneto); Raffaele Paone (ARSSA, Calabria); Costanza Calzolari, Fabrizio Ungaro (CNR Ibimet); Luigi D Acqui

(CNR ISE); Lorenzo Gardin (Consorzio Lamma, Toscana); Edoardo Costantini, Maria Fantappi, Giovanni L Abate, Sergio Pellegrini, (CREA Firenze); Maria Teresa Dell Abate, Rosario Napoli (CREA Roma); Stefano Barbieri (ERSA-FVG); Stefano Brenna (ERSAF, Lombardia); Fabio Petrella, Paolo Martali (IPLA, Piemonte); Michele Munafi, Fiorenzo Fumanti (ISPRA); Amedeo D Antonio (Regione Campania); Paola Tarocco, Francesca Staffilani (Regione Emilia-Romagna SGSS); Stefano Pini (Regione Liguria), Mauro Tiberi (Regione Marche); Luigi Scamarcio (Regione Puglia); Fabio Guaitoli (Regione Sicilia); Claudio Zaccone (Universita di Foggia); Adele Muscolo (Universita Mediterranea di Reggio Calabria).

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: Neural Networks and GLM, according to soil region

Validation: Mean Error (ME) of the prediction is 1.688 Mg/ha, MAE 25.57 Mg/ha, Root Mean Squared Error (RMSE) is 36.24 Mg/ha.

Contact Point:

Data holder: Edoardo Costantini

Institution: Research centre for agriculture and environment (CREA)

Email or website: edoardo.costantini@crea.gov.it

D87: Jamaica

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 77

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2042

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Quantile Regression Forest

Validation: N/A

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D88: Japan

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 6254

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Measured

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling
Mapping method: Geomatching / Class-matching
Mapping method details: N/A
Validation: N/A

Contact Point:

Data holder: Cropland: Hiroshi Obara, Forests: Shigehiro Ishizuka
Institution: Cropland: Institute for Agro-Environmental Sciences, NARO,
Forests: Forestry and Forest Products Research Institute
Email or website: Cropland: obara@affrc.go.jp, Forest: ishiz03@ffpri.affrc.go.jp

D89: Jordan**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 1072
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Measured
Coarse fragments method: Measured
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Regression Kriging
Mapping method details: Multiple linear regression, Kriging
Validation: R² (0.92)

Contact Point:

Data holder: Mahmoud alfraihat

Institution: MOA

Email or website: Mahmoudalfrehat@gmail.com

D90: Kazakhstan**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 502

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Afghanistan, Pakistan, Turkmenistan, Tajikistan, Kyrgyzstan and the original data provided by Kazakhstan and Uzbekistan.

Validation: $R^2 = 0.88$, RMSE=10.5 t/ha

Contact Point:

Data holder: Maira Kussainova

Institution: Kazakh Research Institute of Soil Science and Agrochemistry

named after U.U. Uspanov

Email or website: madgu@inbox.ru

D91: Kenya

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 2059

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Anderson, J. M. and J. S. I. Ingram (eds). 1993. Tropical Soil Biology and Fertility. A handbook of Methods. C.A.B. International, Wallingford, UK.

Hinga G., Muchena F.N. and Njihia C.M., (ed 1980). Physical and Chemical methods of soil analysis. National Agricultural Laboratories, Nairobi
Zhu A.X., Liud J., Dud f., Zhang S.J., Qin C.Z., Burtd J., Behrense T., and Scholtene T. Predictive soil mapping with limited sample data. European Journal of Soil Science, May 2015, 66, 535-547.

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Other

Mapping method details: Environmental Correlation

Validation: RMSE=39.72 t/ha ;AC=0.657806; MAE=28.958 t/ha; SDOV=36.117432; ME=1.09 t/ha Where: RMSE is root mean square error; AC is agreement coefficient; MAE is mean absolute error; SDOV is standard deviation of observed values; ME is mean error.

Contact Point:

Data holder: Peter Kamoni, Matolo Nyamai

Institution: Kenya Agricultural and Livestock Research Organization

Email or website: Peter kamoni(pkamoni@gmail.com) Matolo Nyamai(matolonyamai@gmail.com/nyamai.matolo@kalro.org)

D92: Kiribati**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D93: Kuwait**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2044

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D94: Kyrgyzstan

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2043

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Afghanistan, Pakistan, Turkmenistan, Tajikistan, Kyrgyzstan and the original data provided by Kazakhstan and Uzbekistan.

Validation: $R^2 = 0.88$, RMSE=10.5 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D95: Lao People's Democratic Republic

GSOCmap layer source:

Map source: Joint effort with GSP

Data specifications:

Number of samples: 155

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: $R^2=0.34$; $MAE=13.65$; $RMSE=21.44$.

Contact Point:

Data holder: Ministry of Agriculture and Forestry of Laos

Institution: Ministry of Agriculture and Forestry of Laos

Email or website: <http://www.maf.gov.la/>

D96: Latvia**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 349

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LU-CAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D97: Lebanon**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 450

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Estimated

Citation: Darwish T., Khawlie M., Jomaa M., Awad M. Abou Daher and P. Zdruli (2002). A survey to upgrade information for soil mapping and management in Lebanon. Options Mediterraneennes, Series A: Mediterranean Seminars, number 50: 57-71.

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Other

Mapping method details: Kriging

Validation: N/A

Contact Point:

Data holder: Talal Darwish

Institution: Handler Institute: National Council for Scientific Research CNRS
Lebanon

Email or website: tdarwich@cnrs.edu.lb; tlldarwish@gmail.com

D98: Lesotho**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 74

Time frame: before 1990

Soil Organic Carbon method: N/A

Bulk density method: Measured

Coarse fragments method: Measured

Citation: Cauley, P. M., 1986. Benchmark soils of Lesotho: their classification, interpretation, use, and management. Maseru, Lesotho.

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: Multiple linear regression, R Kriging, Random forest
Validation: N/A

Contact Point:

Data holder: Koetlisi Koetlisi

Institution: Ministry of Forestry, Range and Soil Conservation

Email or website: koetlisika@email.com and lesis2017@gmail.com

D99: Liberia**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 48

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2045

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya,

Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D100: Libya

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 14

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2046

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon,

Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen
Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D101: Lithuania**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 356

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LU-CAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D102: Luxembourg

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 3492

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Linear Models

Mapping method details: Generalized Additive Models (covariates: Land use, Elevation, precipitation, temperature, C factor, % Clay (80x80m))

Validation: For cropland soils: R² = 0.66, RMSE = 5.5 g C kg⁻¹

Contact Point:

Data holder: Marx Simone

Institution: Administration of agricultural technical services - Soil department

Email or website: simone.marx@asta.etat.lu

D103: Madagascar**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 1193

Time frame: after 1990

Soil Organic Carbon method: mixed

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Ramifehiarivo, N., et al., Mapping soil organic carbon on a national scale: Towards an improved and updated map of Madagascar, (2016), <http://dx.doi.org/10.1016/j.geodrs.2016.12.002>

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: R²=0.59; RMSE=25.8 t/ha

Contact Point:

Data holder: Andry Andriamananjara

Institution: Laboratoire des Radioisotopes, University of Antananarivo

Email or website: njaraandry1@gmail.com

D104: Malawi

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 4922

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Linear Models

Mapping method details: Multiple linear regression

Validation: N/A

Contact Point:

Data holder: Kefasi Kamoyo/ John Mussa

Institution: Department of Land Resources Conservation, Ministry of Agriculture, Irrigation and Water Development,

Email or website: kamokefa@yahoo.com/ mussajj@gmail.com

D105: Malaysia

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D106: Maldives**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D107: Mali**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 667

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2048

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D108: Malta

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 19

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D109: Marshall Islands

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D110: Mauritania

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 11

Time frame: before 1990

Soil Organic Carbon method: N/A

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2049

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D111: Mauritius

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D112: Mexico

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 36015

Time frame: before and after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: mixed

Coarse fragments method: Measured

Citation: Soil Organic Carbon Map 2017. Mexico.

Methodological specifications:

Mapping technique: Conventional upscaling, Digital soil mapping

Mapping method: Mixed

Mapping method details: Mixed. Iterative Calibration map and Linear Regression. To spatially represent SOC information we use regionalization models that represent changes in average SOC values as a function of changes in factors of soil formation (or loss); such as gravity (inclination), climate, vegetation, land use, water erosion, deforestation, degradation and recovery. Each logical relationship has both exception rules and quantitative trend graphs for the continuous range of carbon values, which is established from the available cartographic information. Digital soil mapping techniques are also used to build statistical models and spatial predictions of SOC and depth. The ongoing development and the implementation of a national soil spatial inference engine assisted with high performance computing techniques will allow to periodically provide wall-to-wall SOC estimates at relevant scales for natural resources management.

Validation: Four error factors are considered: (1) Disaggregation or level of detail in the available data. (2) Density or number of field observations per study surface. (3) Dispersion or heterogeneity represented by the coefficient of variation (Cv) obtained from the quotient of the standard deviation and the mean of each covariate of organic carbon (relief, geology, climate, vegetation, human management and various soil processes), and (4) Representation or congruence (qualitative evaluation of expert) between the study sites (points) and the carbon magnitude polygons represented. For the estimation of uncertainties follows the good practices suggestions from the IPCC (2003) through the use of the inverse of variance and R².

Contact Point:

Data holder: Carlos Cruz-Gaistardo

Institution: 1 Instituto Nacional de Estadística y Geografía. INEGI. omar.cruz@inegi.org.mx 2 Comisión Nacional Forestal. CONAFOR. 3 Instituto Nacional de Investigaciones Forestales Agrícolas y Pecuarias. INIFAP. 4 Red Nacional de Laboratorios para el Análisis, Uso, Conservación y Manejo del Suelo. REDLABs. 5 Colegio de Postgraduados. COLPOS. 6 Colegio de la Frontera Sur. ECOSUR. 7 Organización de las Naciones Unidas para la Agricultura y la Alimentación. FAO. 8 University of Delaware, UDEL. 9 Programa de las Naciones Unidas para el Desarrollo. PNUD.

Email or website: omar.cruz@inegi.org.mx

D113: Micronesia (Federated States of)

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D114: Monaco

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: N/A

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LU-CAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D115: Mongolia**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 512

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model

Validation: ME= 1.0 t/ha, MAE=22.6 t/ha, RMSE=30.0 t/ha, R2=0.36

Contact Point:

Data holder: Bayarsukh Noov

Institution: Institute of Plant and Agriculture Science

Email or website: bayar67@yahoo.com

D116: Montenegro**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D117: Morocco**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 24000

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling, Digital soil mapping

Mapping method: Mixed

Mapping method details: esoter approach for soil unit and idw for regression

Validation: N/A

Contact Point:

Data holder: Dr Rachid Moussadek

Institution: INRA Morocco

Email or website: rachidmoussadek@yahoo.fr

D118: Mozambique**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 2427

Time frame: before and after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest Method

Validation: ME=4.4, R2=0.31

Contact Point:

Data holder: Orlando Inacio Jalane
Institution: Mozambique Agrarian Reserch Institute
Email or website: ojalane@gmail.com

D119: Myanmar**GSOCmap layer source:**

Map source: Joint effort with GSP

Data specifications:

Number of samples: 115
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Pedotransfer functions
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Support vector machines
Mapping method details: Support vector machines
Validation: N/A

Contact Point:

Data holder: N/A
Institution: Ministry of Agriculture, Livestock and Irrigation (MoALI)
Email or website: Su Su Win susuwinmyanmar@gmail.com

D120: Namibia

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 56

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2050

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D121: Nauru

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D122: Nepal

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 6000

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Measured

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Soil Science Division
Institution: Nepal Agricultural Research Council
Email or website: matobigyan@gmail.com

D123: Netherlands**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 21210
Time frame: after 1990
Soil Organic Carbon method: dry combustion
Bulk density method: Pedotransfer functions
Coarse fragments method: No data
Citation: Hoogland, T. D. J. Brus, and D.J.J. Walvoort, (2017, in prep.). 3D-geostatistical interpolation of soil organic matter in the Netherlands.

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Regression Kriging
Mapping method details: R Kriging
Validation: N/A

Contact Point:

Data holder: Dennis Walvoort and Tom Hoogland
Institution: Wageningen University and Research, The Netherlands
Email or website: <https://www.wur.nl/en.htm>

D124: New Zealand

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 2050

Time frame: before and after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Measured

Coarse fragments method: Estimated

Citation: McNeill S., Golubiewski N., Barringer J. (2014): Development and calibration of a soil carbon inventory model for New Zealand. Soil Research 52: 789-804; <http://dx.doi.org/10.1071/SR14020>

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Linear Models

Mapping method details: generalized linear model (GLM)

Validation: A test of the measured and predicted soil carbon stocks using the LENZ level 4 environmental classification model (the best of the four developed) indicates a residual standard error of 24.4 t/ha using a robust Gaussian fit of the model residuals.

Contact Point:

Data holder: Stephen McNeill

Institution: Landcare Research

Email or website: mcneills@landcareresearch.co.nz

D125: Nicaragua

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 4000

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Random forest and Support vector machine

Validation: N/A

Contact Point:

Data holder: Fernando J Mendoza Jara

Institution: Universidad Nacional Agraria

Email or website: fmendoza@ci.una.edu.ni

D126: Niger

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 478

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2051

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D127: Nigeria**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 5545
Time frame: before and after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Pedotransfer functions
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Linear Models
Mapping method details: Multiple linear regression
Validation: RMSE 18.9 t/ha, R² 0.529, ME 9.18 t/ha

Contact Point:

Data holder: Oshadiya Pekun
Institution: Federal department of Agricultural Lands and Climate Change
Management Service
Email or website: oshadiyapekun@gmail.com

D128: Niue**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information
Email or website: soilgrids.org
Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen,

B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D129: North Macedonia

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 4300

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: AVE=0.42; MAE=19.49; RMSE=25.73.

Contact Point:

Data holder: Global Soil Partnership

Institution: Dusko Mukaetov

Email or website: NA

D130: Norway

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 3218

Time frame: before and after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2052

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: ensemble of 6 models: random forest, cubist, kernels, decision trees, principal components and partial least square regression

Validation: N/A

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D131: Oman

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 9

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2053

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D132: Pakistan

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 337

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2054

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Afghanistan, Pakistan, Turkmenistan, Tajikistan, Kyrgyzstan and the original data provided by Kazakhstan and Uzbekistan.

Validation: $R^2 = 0.88$, RMSE=10.5 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D133: Palau

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D135: Panama**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 837

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Regression Kriging

Mapping method details: R Kriging

Validation: N/A

Contact Point:

Data holder: Ivan Ramos

Institution: IDIAP

Email or website: iarz1103@gmail.com

D136: Papua New Guinea**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D137: Paraguay**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 2768

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: ME: 16.6 t/ha; MAE: 33.9; t/ha; RMSE: 31.2 t/ha; R2: 0.1416

Contact Point:

Data holder: Minisry of Agriculture MAG

Institution: Secretary of Environment SEAM

Email or website: no available

D138: Peru**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 1010

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines

Validation: N/A

Contact Point:

Data holder: Honnan Denis Ponte Saldana

Institution: Ministerio de Agricultura y Riego

Email or website: hponte1410@Gmail.com, hponte@minagri.gob.pe

D139: Philippines**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 500

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Measured

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Other

Mapping method details: Kriging

Validation: RMSE = 0.48

Contact Point:

Data holder: Baldwin Morales Pine

Institution: Bureau of Soils and Water Management

Email or website: baldwinmp@gmail.com/rodelcarating@yahoo.com/angelenriquez.bswm@gmail.com

D140: Poland

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 1648

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D141: Portugal

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 476

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D142: Qatar

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2056

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D143: Republic of Korea

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 0

Time frame: external datasets

Soil Organic Carbon method: external datasets

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: external datasets

Citation: N/A

Methodological specifications:

Mapping technique: external datasets

Mapping method: external datasets

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Global Soil Partnership

Institution: N/A

Email or website: N/A

D144: Republic of Moldova

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 0
Time frame: before 1990
Soil Organic Carbon method: estimated
Bulk density method: Pedotransfer functions
Coarse fragments method: No data
Citation: NA

Methodological specifications:

Mapping technique: Conventional upscaling
Mapping method: Other
Mapping method details: N/A
Validation: N/A

Contact Point:

Data holder: Iurii Rozloga
Institution: Institute of Pedology, Agrochemistry and Soil Protection “N. Dimo”, Moldova
Email or website: iu.rozloga@gmail.com

D145: Romania**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 1384
Time frame: after 1990
Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D146: Russian Federation

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 150000

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Estimated

Citation: Golozubov O.M., Chernova O.V. Using multi-scale old and modern maps combined with current soil monitoring data for online mapping the soil organic carbon stocks // International Conference “Global Soil Map 2017” Moscow, Russia, July 4-6, 2017. Materials of conference. p. 36.

Methodological specifications:

Mapping technique: Conventional upscaling, Digital soil mapping

Mapping method: Mixed

Mapping method details: Used both conventional upscaling and DSM. For DSM: regression equations, IWD and kriging, fuzzy sets depending on area of country and sample density

Validation: Under development yet

Contact Point:

Data holder: Oleg Golozubov

Institution: Soil science department of M.V.Lomonosov MSU, soil data center; Soil science department of Southern State University; Agrochemical data center “Rostovsky” of the Ministry of agriculture of the Russian Federation; Agrochemical data center “Belgorodsky” of the Ministry of agriculture of the Russian Federation; Severtsov Institute of Ecology and Evolution of the Russian Academy of Sciences; Mytishchi filial of Bauman Moscow State Technical University, soil science department; International Institute for Applied Systems Analysis, Ecosystem Services and Management program

Email or website: oleggolozubov@gmail.com

D147: Rwanda

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 88

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2057

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Burundi, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Rwanda, Uganda

Validation: MAE=16 t/ha; RMSE=19.1 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D148: Saint Kitts and Nevis**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D149: Saint Lucia**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D150: Saint Vincent and the Grenadines**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen,

B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D151: Samoa

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D153: Sao Tome e Principe

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D154: Saudi Arabia

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2058

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D155: Senegal

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 678

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: External Datasets (SoilGrids, HWSD)

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Regression Kriging

Mapping method details: R Kriging

Validation: MAE 10.2 t/ha, RMSE 18.1 t/ha, R2: 0,67

Contact Point:

Data holder: Macoumba Loum

Institution: Institut National de Pedologie

Email or website: macoumbaloum@yahoo.fr

D156: Serbia

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D157: Seychelles**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D158: Sierra Leone**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 11

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2059

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D159: Singapore

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D160: Slovakia

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 16748

Time frame: before and after 1990

Soil Organic Carbon method: mixed

Bulk density method: mixed

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Rastislav Skalsky (agricultural soils); Pavel Pavlenda (forest soils)

Institution: National Agricultural and Food Centre, Soil Science and Conservation Research Institute (agricultural soils); National Forestry Centre, Forestry Research Institute (forest soils)

Email or website: r.skalsky@vupop.sk; pavlenda@nlcsk.org

D161: Slovenia

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 1681

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Borut Vrscaj

Institution: Agricultural institute of Slovenia

Email or website: borut.vrscaj@kis.si

D162: Solomon Islands

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D163: Somalia**GSOCmap layer source:**

Map source: Joint effort with GSP

Data specifications:

Number of samples: 257

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines

Validation: ME=3.2 t/ha; MAE=10.7 t/ha; RMSE=16.3 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Somalia Water and Land Information Management (SWALIM)

Email or website: swalim@fao.org

D164: South Africa

GSOCmap layer source:

Map source: Joint effort with GSP

Data specifications:

Number of samples: 11257

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines

Validation: ME=0.07 t/ha; MAE=19.3 t/ha; RMSE=27.6 t/ha

Contact Point:

Data holder: Dr. Maila

Institution: Soil, Climate and Water (ARC-SCW)

Email or website: scwinfo@arc.gis.za

D165: South Sudan

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2061

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D166: Spain**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 2696

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured

Citation: Toth, G., Jones, A., Montanarella, L. (eds.) 2013. "LUCAS Topsoil Survey. Methodology, data and results. JRC Technical Reports. Luxembourg. Publications Office of the European Union, EUR26102 - Scientific and Technical Research series - ISSN 1831-9424 (online); ISBN 978-92-79-32542-7; doi: 10.2788/97922"

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine model for EU based on LUCAS data

Validation: ME = -5.7 t/ha; MAE=32.9 t/ha; RMSE=45.7 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: European Soil Data Centre (ESDAC)

Email or website: esdac@jrc.ec.europa.eu

D167: Sri Lanka**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 233
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: mixed
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Linear Models
Mapping method details: Multiple linear regression
Validation: N/A

Contact Point:

Data holder: Dr. Ajantha de Silva
Institution: Natural Resources Management Centre, Department of Agriculture
Email or website: ajandes@gmail.com

D168: Sudan**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 1584
Time frame: before and after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: mixed

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: STATISTICS_MAXIMUM=4.8131771087646 STATISTICS_MEAN=1.17001577735

STATISTICS_MINIMUM=0.001964766299352 STATISTICS_STDDEV=0.71722792069521

Contact Point:

Data holder: Abdelmagid Ali Elmobarak

Institution: Land and Water Research Centre,

Email or website: melmobarak2012@gmail.com

D169: Suriname

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 178

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2062

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machine

Validation: N/A

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D170: Sweden**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 19097

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: mixed

Coarse fragments method: Measured

Citation: Stendahl, J., Johansson, M.B., Eriksson, E., Nilsson, A., Langvall, O., 2010. Soil Organic Carbon in Swedish Spruce and Pine Forests - Differences in Stock Levels and Regional Patterns. *Silva Fennica* 44, 5-21. <https://doi.org/10.14214/sf.159>

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Other

Mapping method details: Multivariate adaptive regression splines (MAR-Splines)

Validation: N/A

Contact Point:

Data holder: Johan Stendahl

Institution: Dep of Soil and Environment, SLU

Email or website: johan.stendahl@slu.se

D171: Switzerland

GSOCmap layer source:

Map source: Joint effort with GSP

Data specifications:

Number of samples: 1175

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Measured

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: Caret ensemble of Random forest, Cubist, KNN, Bayesian trees, Partial Least Squares Regression, Principal Components Regression

Validation: ME=-7.1 t/ha; MAE=23.1 t/ha; RMSE=34.4 t/ha

Contact Point:

Data holder: Fabio Wegmann

Institution: Federal Office for the Environment, Soil and Biotechnology Division

Email or website: fabio.wegmann@bafu.admin.ch

D172: Syrian Arab Republic**GSOCmap layer source:**

Map source: Joint effort with GSP

Data specifications:

Number of samples: 1220

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines

Validation: ME=-0.03 t/ha; MAE=10.1 t/ha; RMSE=14.3 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: General Commission for Scientific Agriculture Research

Email or website: Syrian Arab Republic, Damascus, Al-Hyjazz Square, General Commission for Scientific Agriculture Research

D173: Tajikistan

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 21

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2065

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Afghanistan, Pakistan, Turkmenistan, Tajikistan, Kyrgyzstan and the original data provided by Kazakhstan and Uzbekistan.

Validation: $R^2 = 0.88$, RMSE=10.5 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D174: United Republic of Tanzania

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 3215

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured

Citation: Kempen, B. 2016. Development of a soil carbon map for the United Republic of Tanzania. ISRIC, Wageningen.

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Regression Kriging

Mapping method details: R. Kriging, Random forest

Validation: ME: -0.00, MAE: 12.3, RMSE: 18.0, R2: 0.53

Contact Point:

Data holder: Joseph D. Mbogoni

Institution: Agricultural Research Institute Mlingano

Email or website: jdjmbogoni@gmail.com

D175: Thailand

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 70000
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: mixed
Coarse fragments method: Estimated
Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling
Mapping method: Geomatching / Class-matching
Mapping method details: N/A
Validation: N/A

Contact Point:

Data holder: Suradesh Tiewtrakool
Institution: Land Development Department
Email or website: dglidd@ldd.go.th

D176: Timor-Leste**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information
Email or website: soilgrids.org
Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D177: Togo

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 9

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2064

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D178: Tokelau

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D179: Tonga

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D180: Trinidad and Tobago

GSOCmap layer source:

Map source: Joint effort with GSP

Data specifications:

Number of samples: 122
Time frame: before 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: mixed
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Geomatching / Class-matching
Mapping method details: Support vector machine based on statistical simulation of the position of sampling points
Validation: N/A

Contact Point:

Data holder: Ronald Roopnarine. Gaius Eudoxie
Institution: University of The West Indies
Email or website: ronald.roopnarine@sta.uwi.edu

D181: Tunisia**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 58
Time frame: before and after 1990
Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2067

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D182: Turkey

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 31800

Time frame: after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: Measured

Coarse fragments method: No data

Citation: within the context of “Establishment of National Geospatial Soil Fertility and Soil Information System Project UTF/TUR/057/TUR, in collaboration with FAO

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Regression Kriging

Mapping method details: Multiple Regression Kriging

Validation: AVE=0.51; MAE=13.36; RMSE=20.84.

Contact Point:

Data holder: Dr. Bulent Sonmez, Doc. Dr. Aynur Ozbance

Institution: General Directorate of Agricultural Research And Policies, Soil, Fertilizer and Water Resources Central Research Institute Ankara, TURKEY

Email or website: bulent.sonmez@tarim.gov.tr, aynur.ozbahce@tarim.gov.tr

D183: Turkmenistan

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: N/A

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T.,

and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2066

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Afghanistan, Pakistan, Turkmenistan, Tajikistan, Kyrgyzstan and the original data provided by Kazakhstan and Uzbekistan.

Validation: $R^2 = 0.88$, RMSE=10.5 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D184: Tuvalu

GSOCmap layer source:

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information

Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. Soil, 7(1), 217-240.

D185: Uganda

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 12

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2068

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Burundi, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Rwanda, Uganda

Validation: MAE=16 t/ha; RMSE=19.1 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D186: Ukraine

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 3931

Time frame: before and after 1990

Soil Organic Carbon method: mixed

Bulk density method: mixed

Coarse fragments method: No data

Citation: K. Viatkin, Yu. Zalavskyi, O. Bihun, V. Lebed, O. Sherstiuk, I. Plisko, S. Nakisko. Creation of the Ukrainian national soil organic carbon stocks map using digital soil mapping methods // Soil Science and Agrochemistry, Almaty, Kazakhstan, n2 2018, p. 5-17

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Mixed

Mapping method details: Random forest

Validation: For mineral soils: ME=0.1 t/ha, MAE=13.2 t/ha; RMSE=18.2 t/ha, R2=0.56. For peat soils: ME=-1.2 t/ha, MAE=18.7 t/ha, RMSE=24.5 t/ha, R2=0.22.

Contact Point:

Data holder: Sviatoslav Baliuk

Institution: National Scientific Center "Institute for Soil Science and Agrochemistry Research named after O.N. Sokolovsky" (NSC ISSAR) and contributors

Email or website: pochva@meta.ua

D187: United Arab Emirates

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 0

Time frame: NA

Soil Organic Carbon method: N/A

Bulk density method: N/A

Coarse fragments method: No data

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2018

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D188: United Kingdom of Great Britain and Northern Ireland

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 17421

Time frame: before and after 1990

Soil Organic Carbon method: mixed

Bulk density method: Pedotransfer functions

Coarse fragments method: Estimated

Citation: Bradley, R.I., Milne, R., Bell, J., Lilly, A., Jordan, C. and Higgins, A. (2005), A soil carbon and land use database for the United Kingdom. Soil Use and Management, 21: 363-369. doi:[10.1079/SUM2005351](https://doi.org/10.1079/SUM2005351)

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Caroline Keay

Institution: Cranfield University

Email or website: c.keay@cranfield.ac.uk

D189: United States of America

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 10000

Time frame: before and after 1990

Soil Organic Carbon method: dry combustion

Bulk density method: mixed

Coarse fragments method: Estimated

Citation: N/A

Methodological specifications:

Mapping technique: Conventional upscaling

Mapping method: Geomatching / Class-matching

Mapping method details: N/A

Validation: N/A

Contact Point:

Data holder: Micheal Robotham

Institution: Natural Resource Conservation Service

Email or website: michael.robatham@wdc.usda.gov

D190: Uruguay**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 160

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Estimated

Citation: Direccion General de Recursos Naturales-DGRN Ministerio de Ganaderia y Agricultura y Pesca-MGAP - Uruguay 2017

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Regression Kriging

Mapping method details: Multiple linear regression, R Kriging,

Validation: RMSE = 0.4566, MAE= 0.3558, me_mean= -0.0002158, R2=0.5549

Contact Point:

Data holder: Martin Dell Acqua Gonzalo Pereira Pablo Prieto Fernando Fontes Fabian Davila

Institution: Direccion General de Recursos Naturales

Email or website: mdellacqua@mgap.gub.uy mdavila@mgap.gub.uy ffontes@mgap.gub.uy pprieto@mgap.gub.uy gpereira@mgap.gub.uy

D191: Uzbekistan**GSOCmap layer source:**

Map source: Country submission

Data specifications:

Number of samples: 4969
Time frame: after 1990
Soil Organic Carbon method: wet oxidation
Bulk density method: Measured
Coarse fragments method: No data
Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping
Mapping method: Random forest
Mapping method details: Random forest model based on the ensemble globally available data from Afghanistan, Pakistan, Turkmenistan, Tajikistan, Kyrgyzstan and the original data provided by Kazakhstan and Uzbekistan.
Validation: $R^2 = 0.88$, RMSE=10.5 t/ha

Contact Point:

Data holder: Bakhodir Ruziboev
Institution: UZGIP Design and Research Institute
Email or website: uzgip_tas@umail.uz, uzgip@bk.ru

D192: Vanuatu**GSOCmap layer source:**

Map source: External (soilgrids.org)

Details:

Institution: ISRIC World Soil Information
Email or website: soilgrids.org

Citation: Poggio, L., De Sousa, L. M., Batjes, N. H., Heuvelink, G., Kempen, B., Ribeiro, E., & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *Soil*, 7(1), 217-240.

D193: Venezuela (Bolivarian Republic of)

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 310

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: No data

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: RMSE = 41.1 t/ha, R^2 = 0.0272, Mean error = 12.8 t/ha, Mean absolute error = 19.6 t/ha

Contact Point:

Data holder: Juan C.Rey

Institution: Sociedad Venezolana de la Ciencia del Suelo (SVCS)

Email or website: svcs.org

D194: Viet Nam

GSOCmap layer source:

Map source: Country submission

Data specifications:

Number of samples: 1024

Time frame: after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Measured

Coarse fragments method: Measured

Citation: N/A

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest

Validation: $R^2=0.34$; $MAE=13.65$; $RMSE=21.44$.

Contact Point:

Data holder: Vu Manh Quyet

Institution: Soils and Fertilizers Research Institute

Email or website: quyetvm.sfri@mard.gov.vn; vmquyet@gmail.com

D195: Yemen

GSOCmap layer source:

Map source: GSP gap-filling

Data specifications:

Number of samples: 270

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: Pedotransfer functions

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2069

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Support vector machines

Mapping method details: Support vector machines model based on the ensemble globally available data from Algeria, Chad, Egypt, Iraq, Jordan, Lebanon, Libya, Mali, Mauritania, Morocco, Niger, Oman, Qatar, Saudi Arabia, Sudan, Syrian Arab Republic, United Arab Emirates, Western Sahara and Yemen

Validation: MAE=6.8 t/ha; RMSE = 10.94 t/ha

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D196: Zambia**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 460

Time frame: before 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2070

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

D197: Zimbabwe**GSOCmap layer source:**

Map source: GSP gap-filling

Data specifications:

Number of samples: 179

Time frame: before and after 1990

Soil Organic Carbon method: wet oxidation

Bulk density method: mixed

Coarse fragments method: Measured/Estimated

Citation: Batjes, N. H., Ribeiro, E., van Oostrum, A., Leenaars, J., Hengl, T., and Mendes de Jesus, J.: WoSIS: providing standardised soil profile data for the world, Earth Syst. Sci. Data, 9, 1-14, <https://doi.org/10.5194/essd-9-1-2017>, 2071

Methodological specifications:

Mapping technique: Digital soil mapping

Mapping method: Random forest

Mapping method details: Random forest model based on the ensemble globally available data from Angola, Botswana, Cameroon, Congo, Gabon, Guinea, Burundi, Central African Republic, Democratic Republic of the Congo, Kenya, Malawi, Namibia, South Africa, Rwanda, United Republic of Tanzania, Uganda, Zambia, Zimbabwe

Validation: MAE=27 t/ha; RMSE=43.7 t/ha; R²=0.19.

Contact Point:

Data holder: Global Soil Partnership

Institution: Global Soil Partnership

Email or website: <http://www.fao.org/global-soil-partnership/en/>

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